



UX Series DC Charging Piles

UI Operation Manual

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1.Main Interface

1.1.Overview of the Main Interface

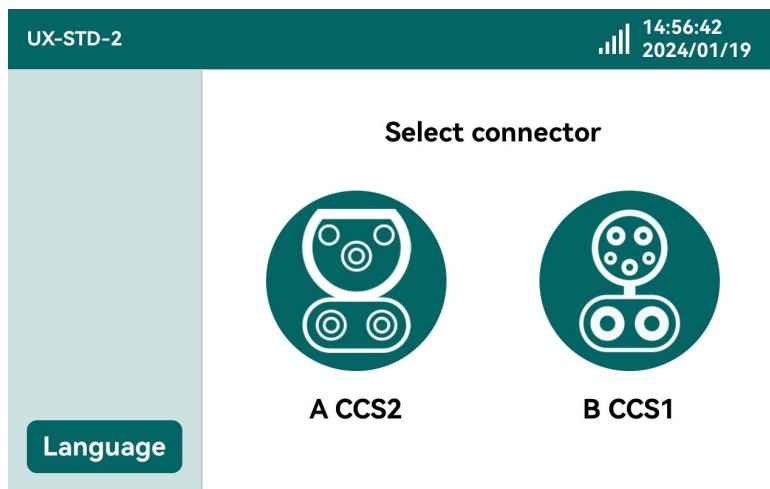


图 1-1 主界面

The main interface of the UX DC charging pile can be divided into three parts:

- Top information bar
- Left side function bar
- Right-side charging operation window

1) Top information bar

- a. **UX-STD-2** Assign a number to the charging station, and the identification number for connecting to the OCPP platform;
- b. network status display,  Indicates successful connection with the platform,  Indicates disconnection from the platform.

2) Left function bar: Language switch (language);

3) Right-side charging operation window:

- a. Operating window for the charging process:

-  European standard charging option;
-  Target charging option; (not shown in Figure 1-1)
-  American Standard charging options.

- b. Users can perform the start/stop charging operation and view the charging-related information in this window. When the charging gun malfunctions, a red fault code will be

displayed in the corresponding charging port process interface, for example: **1003**.

1.1.1. Instructions for Language Switching Operation

1) Click the "Language" button on the main interface to switch languages.

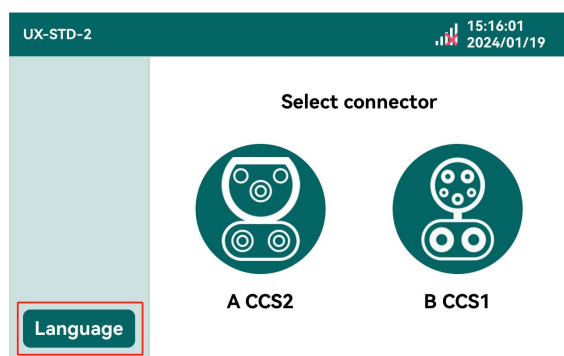
Language

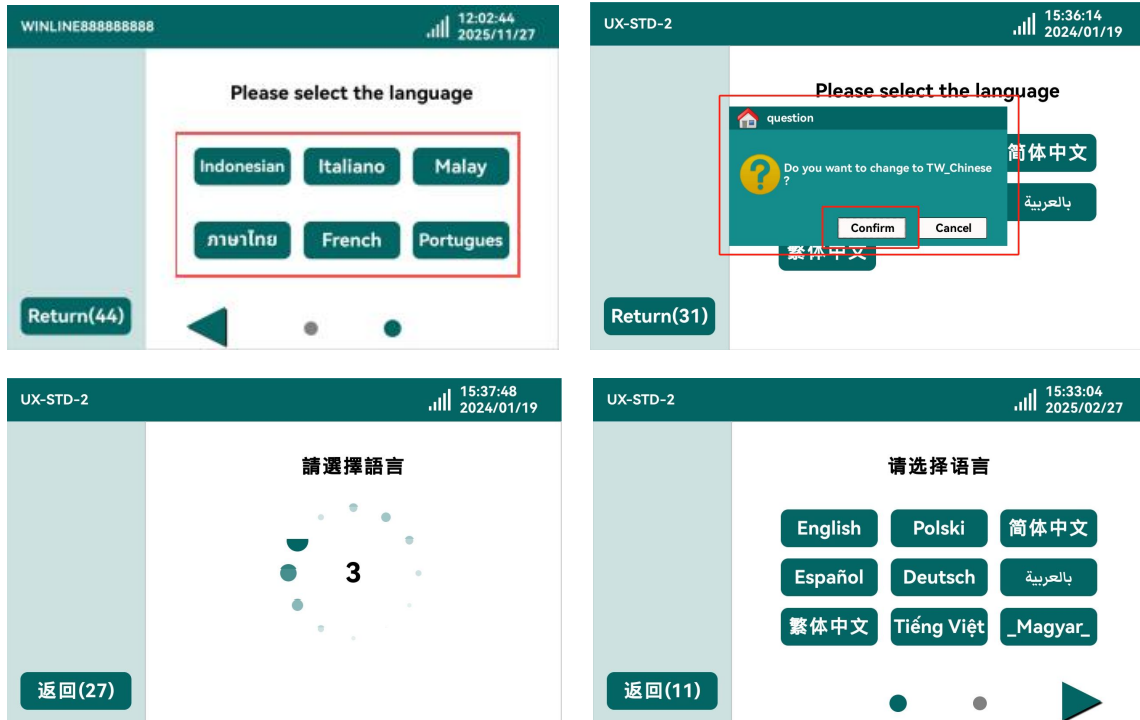
2) Select the language you need to switch to:

English	英语
German	德语
Polish	波兰语
Chinese	简体中文
Spanish	西班牙语
Arabic	阿拉伯语
Traditional	繁体中文
Vietnamese	越南语
Magyar	匈牙利语
Indonesian	印度尼西亚语
Italian	意大利语
Malay	马来语
Thai	泰语
French	法语
Portugues	葡萄牙语

3) English, German, Polish, Simplified Chinese, Spanish, Arabic, Traditional Chinese, Vietnamese, Hungarian, Indonesian, Indonesian, Italian, Malay, Thai, French, Portuguese.

4) Click "Confirm"





1.2. Charging Procedure

The interface between the UI and user operation is the charging window on the right side. Different gun ports correspond to different windows, and the operations within each window are independent of each other. Taking the European standard gun as an example, this article introduces the charging-related interface.

The charging interface process is shown in the following figure:

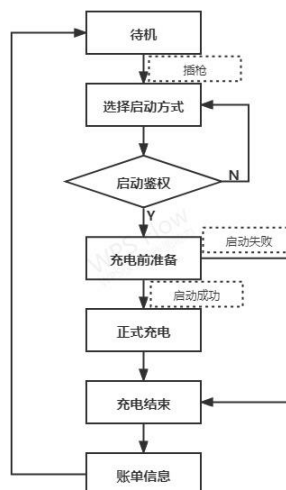


图 1-2 充电界面流程图

1.2.1. Select the startup method

After the charging gun is connected to the vehicle, the corresponding icon of the gun in the

charging interface will have a gray circle added to it. change from  to .

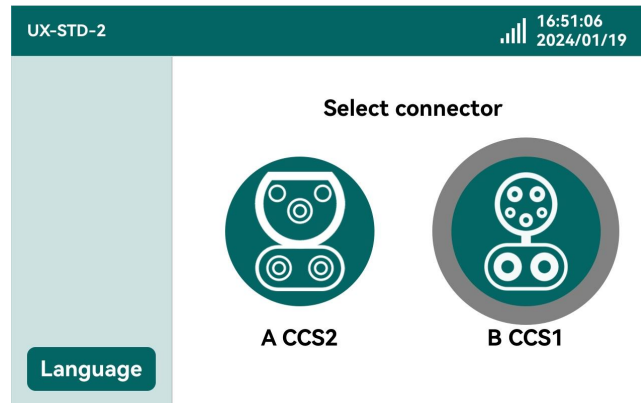


Figure 1-3 The interface showing successful connection of the charging gun to the vehicle

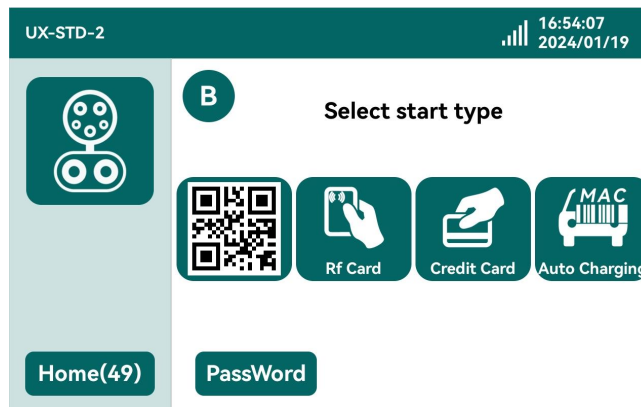


Figure 1-4 Interface for Selecting Startup Mode

As shown in the B gun window in Figures 1-4, there are 5 function buttons on the current page, listed in order as follows: QR code scanning for charging, card swiping charging button (RF Card), credit card charging (POS machine), plug-and-charge, password charging button (PassWord).

- QR code scanning for charging: Scan the code to start the charging process;
- Swipe card charging button: Initiates the charging process by swiping the card;
- Credit card charging button (POS machine): Initiates the credit card charging process;
- Password activation button: Initiates the password-based charging process. Enter the charging password to start the charging.

1.2.2. Five charging mode options

Click on the interface to select the charging mode:

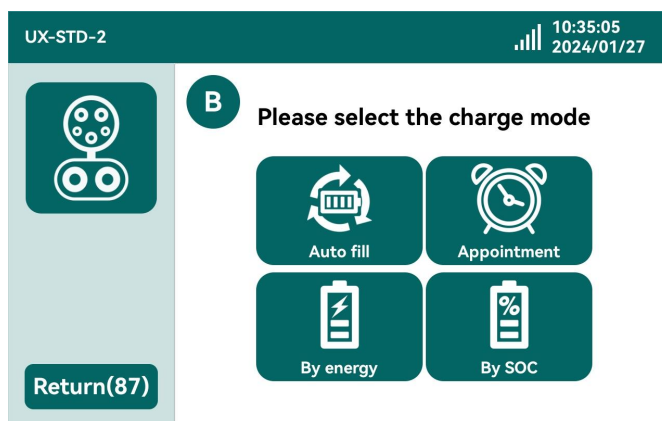


Figure 1-5 Four charging modes

After selecting the card-swipe charging or credit card charging option, the user can choose to click on different charging modes on the main interface. There are four charging modes available for selection:

自动充满模式	定时充电模式	定量充电模式	SOC 充电模式
 Auto fill	 Appointment	 By energy	 By SOC

- 1) **Automatic charging mode:** The charging station communicates with the on-board BMS. The charging process and charging time are entirely controlled by the BMS. After clicking the icon button, you will be directed to the card-swiping authorization page;
- 2) **Scheduled Charging Mode:** Manually set the start and duration of charging times. When the set start time is reached, the charger will automatically start charging. When the set duration of charging is reached, the charger will stop charging. If the scheduled start time is earlier than the current time, the system will default to starting charging at the next day's time; if the set duration is '0', the vehicle's BMS will control the automatic full charging.

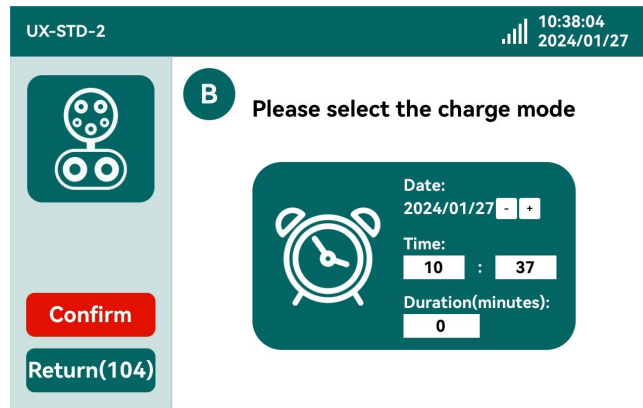


Figure 1-6 Sets the start charging time and the continuous charging time

a. Operation steps: Click on the "Book Charging" icon , Set the start charging time and the continuous charging time , Click "Confirm" , a. Finally, swipe the card to start the charging process.

b. Time specification: The set time is based on the displayed time on the screen.

c. Transactions that are stopped due to reaching the charging duration will display the fault code: 1005 on the screen.

3) Quantitative charging mode: Manually set the charging capacity. Once the charging capacity reaches the preset value, the charging process will automatically stop.

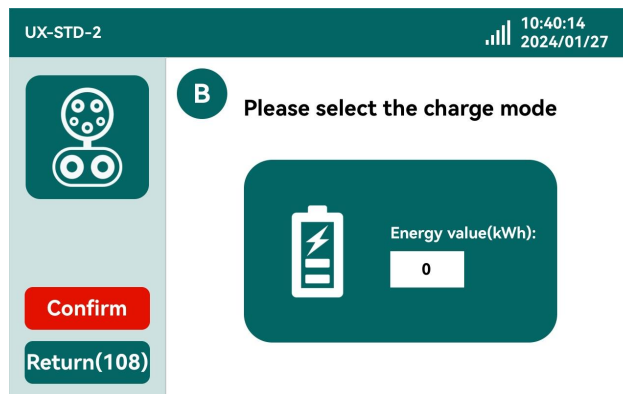


Figure 1-7 Set charging voltage

a. Operation steps: Click on the "Book Charging" icon , Set charging

capacity , Click "Confirm" , Finally, swipe the card to start the charging process.

b. Transactions that are stopped due to reaching the charging capacity are displayed with the fault code: 1006 on the screen.

4) SOC Charging Mode: Manually set the charging SOC. The charging will automatically stop once the vehicle's SOC reaches the preset charging SOC. If the set SOC value is lower than the current actual SOC of the vehicle, the charger will immediately stop charging after it is started.

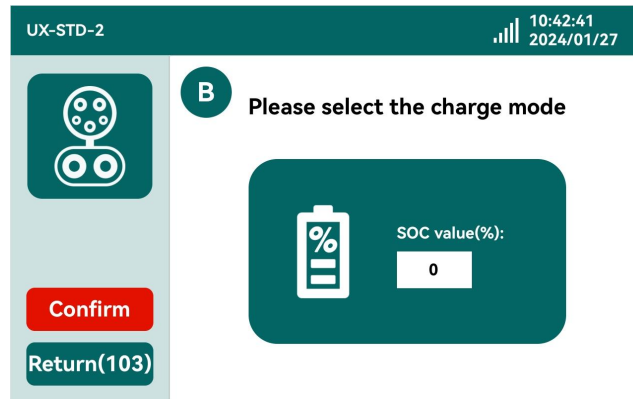
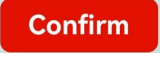


Figure 1-8 Setting the charging SOC

a. Operation steps: Click on the "Book Charging" icon , Set charging SOC , Click "Confirm" , Finally, swipe the card to start the charging process.

b. Transactions that are stopped due to reaching the set SOC are displayed on the screen with the fault code: 9011.

5) The platform initiates the appointment charging mode

The user makes an appointment for the charging station on the platform. Once the appointment is successful, the charging station enters the reserved state. The interface is as follows:

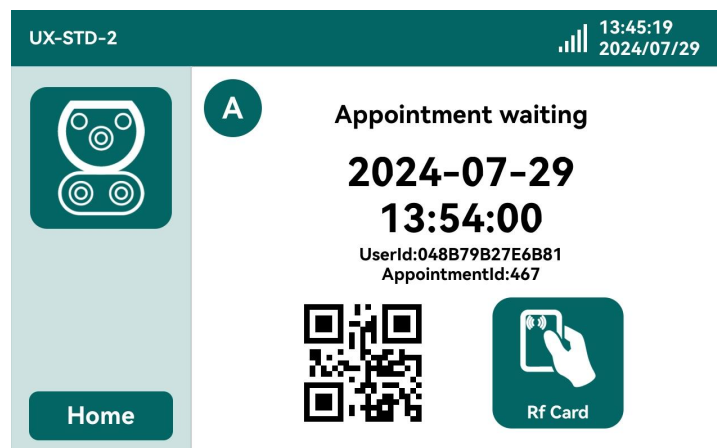


Figure 1-9 Reservation Charging Interface

The information displayed on the interface is as follows: the deadline for this reservation, the user ID of the reservation, and the reservation number. In the reservation state, the user has two options for charging - scanning code or swiping card.

1.2.3.Preparations before charging

After selecting the startup method, the authentication process begins.

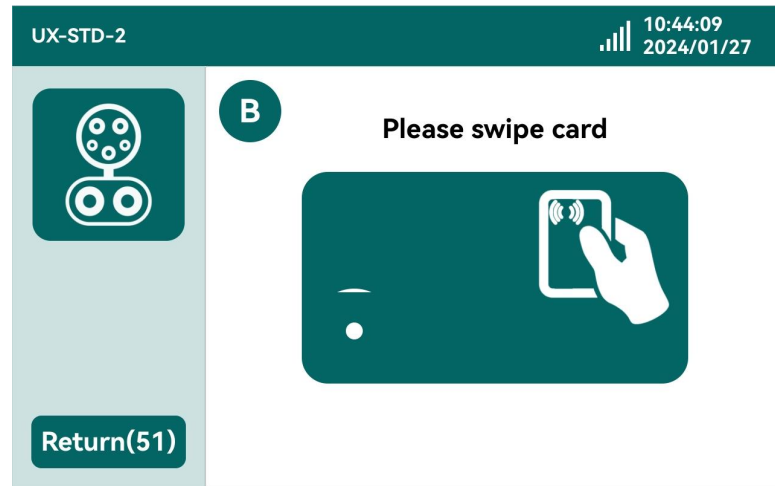


Figure 1-10 Card Authentication Interface

After the authentication process is completed, the charging station enters the preparatory stage for charging. During this stage, the equipment conducts insulation testing and pre-charging operations.

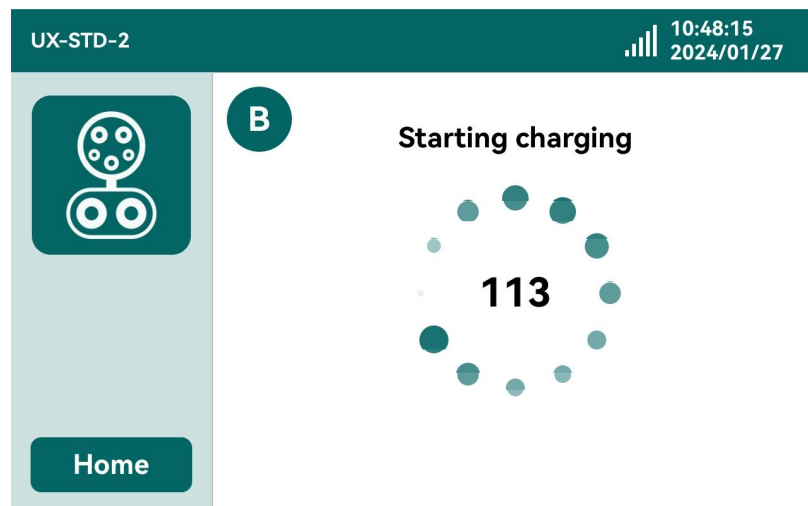


Figure 1-11 Startup Interface

1.2.4.Start the formal charging process

After all the preparatory work before charging is completed, the charging station enters the

formal charging phase.

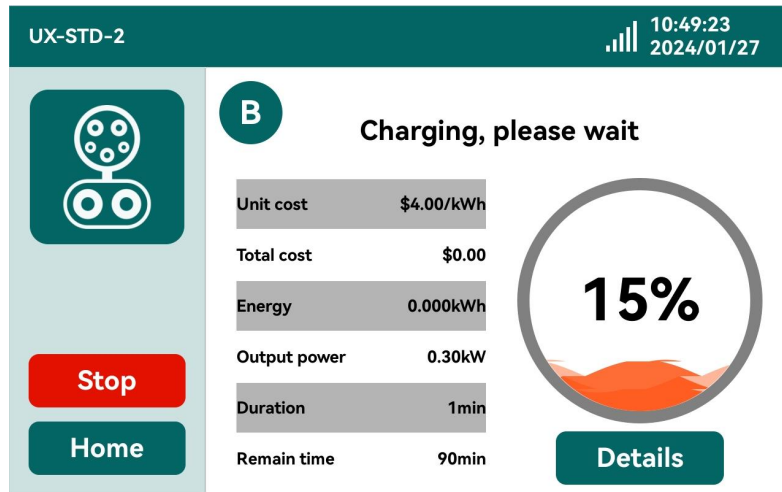


Figure 1-12 Charging Interface

On the current page, users can view the current period's electricity price (Unit cost), the total amount charged (Total cost), the charging power (Energy), the output power, the charging duration (Duration), and the remaining duration (Remain time). Users can click the stop button to do so. , Manually stop charging.

1.2.5.Charging completed

When the vehicle battery is fully charged, the vehicle automatically stops, the user manually stops, the server issues a stop command, or charging cannot continue due to equipment failure, the charging station enters the end charging process.

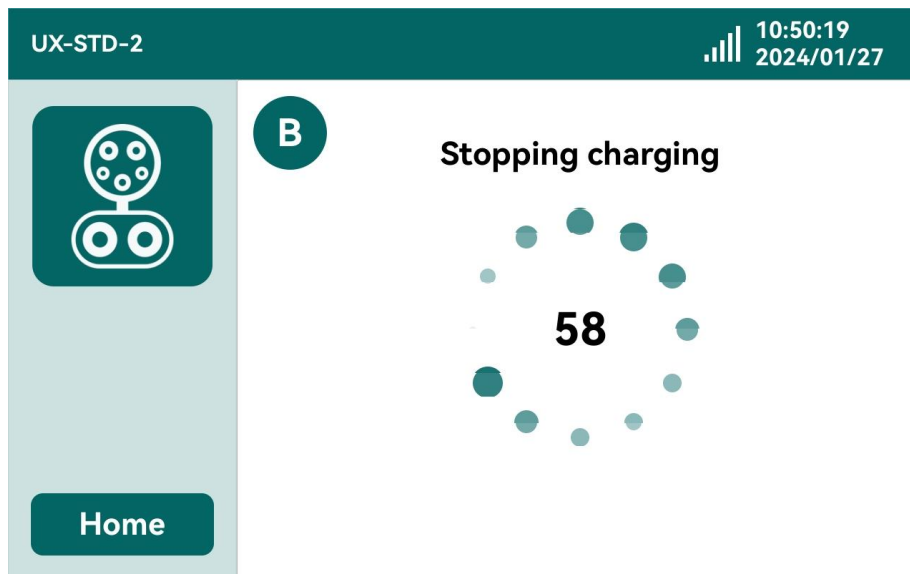


Figure 1-13 Interface for the end of the process

1.2.6. Billing Information

After the charging is completed, if the charging station is connected to the platform server, the charging station will settle this order with the platform and display the bill information to the user; if there is no platform connection, this interface will only show the charging settlement information.

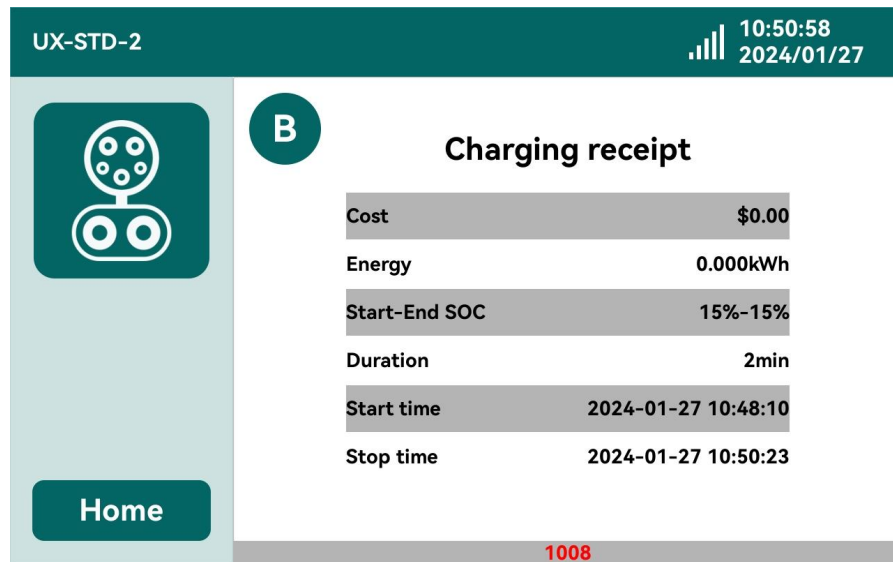


Figure 1-14 Billing Interface

In the current interface, users can view the billing information for this charging session: the total charging amount, the total charging capacity, the SOC from start to end (battery SOC at the end of charging), charging duration, charging start time, charging end time, and the red numerical codes below the bill, which represent the reason for the cessation of this charging session.

Click the "Home" button in the upper right corner of the window.  Exit the bill interface. When the normal charging is completed (by manually clicking the stop button, swiping the card to stop, the battery being fully charged, etc.), when the fault occurs or the startup fails, this interface will be entered. Therefore, you can view the stop reason on this interface and perform maintenance on the device.

1.3.(Startup Method) Operation Procedure for Charging via M1 Card Swipe

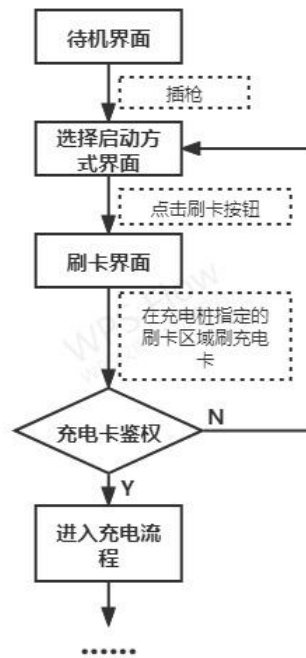
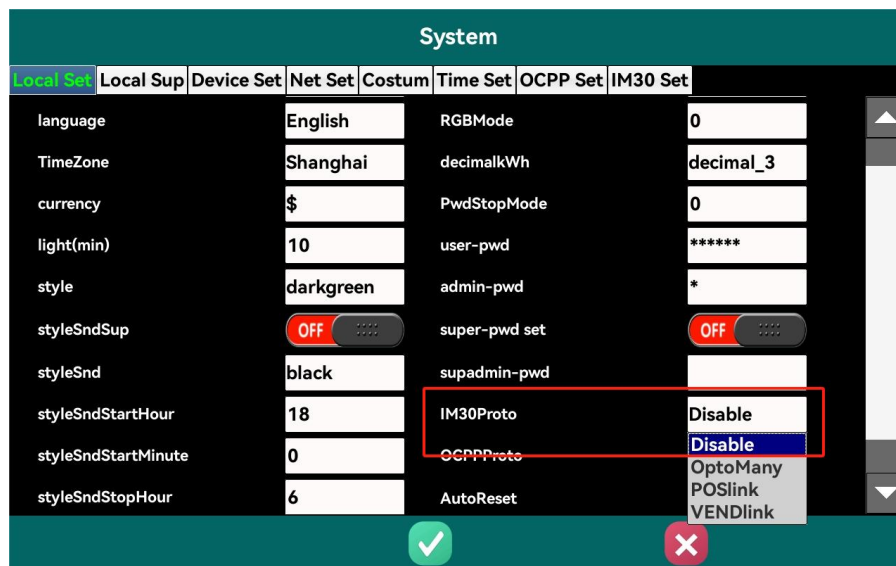


Figure 1-15 Card Swipe Startup Process

The start-up method for the POS machine using card swiping requires the background to be opened. The opening path is as shown in the following figure:



In the IM30Proto options, select the "Disable" mode, and the M1 card startup button will be enabled on the home page. After the user connects the charging gun to the vehicle, if they need to start charging by swiping the card, click the swipe button.  , The page will

navigate to the card-swiping interface. (Note: After 60 seconds on the card-swiping interface, it will automatically return to the interface for selecting the startup method. Only the charging station on the card-swiping interface will perform the card-searching operation.)

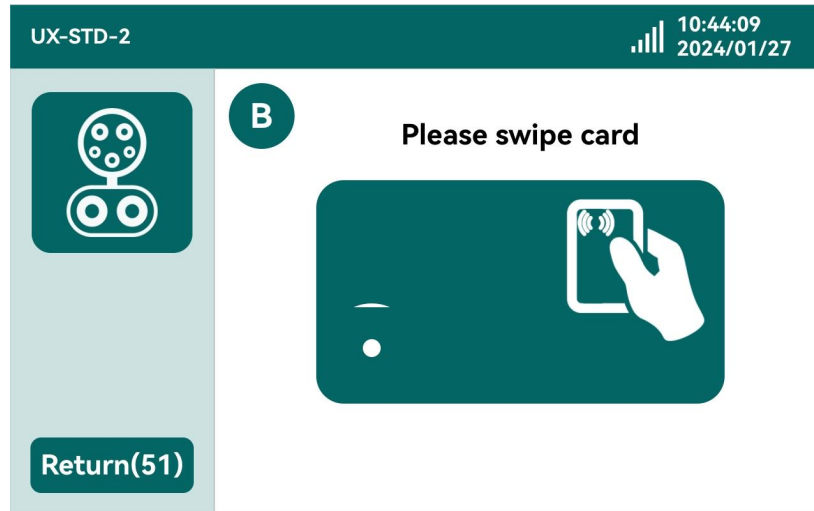


Figure 1-16 Card Swipe Interface

After the user swipes the card and locates the card, the page automatically jumps to the card authentication interface. On this current page, the charging station will conduct card number authentication with the platform. If the authentication is successful, the charging start process will be initiated; if the authentication fails, the interface will return to the selection of start methods interface.

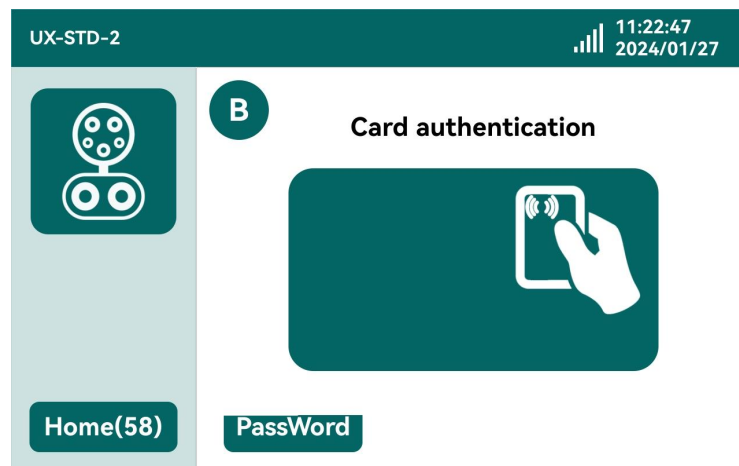


Figure 1-17 Card Authentication Interface

If the authentication is successful, after entering the formal charging process, the charging procedure is as follows: 1.2.4. When the charging is completed, a card swipe is required to stop the process. The charging completion procedure is as described in 1.2.5 and

1.2.6.

1.4.(Startup Method) Credit Card Charging Operation Procedure

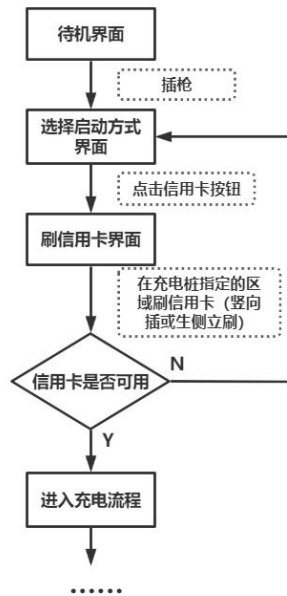


Figure 1-18 Credit Card Activation Process

After the user connects the charging gun to the vehicle, if they need to start the charging

process using a credit card, they should click the credit card button.



, Navigate to the credit card payment interface.

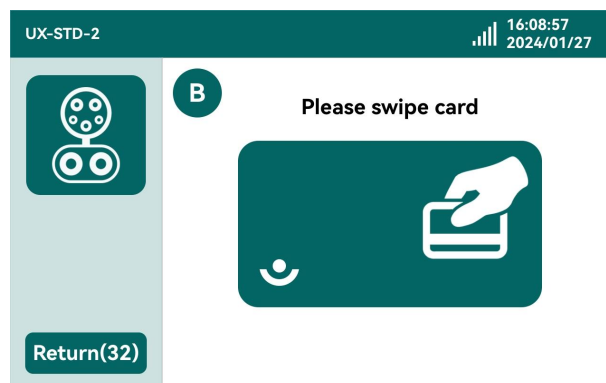


Figure 1-19 Credit Card Swipe Interface

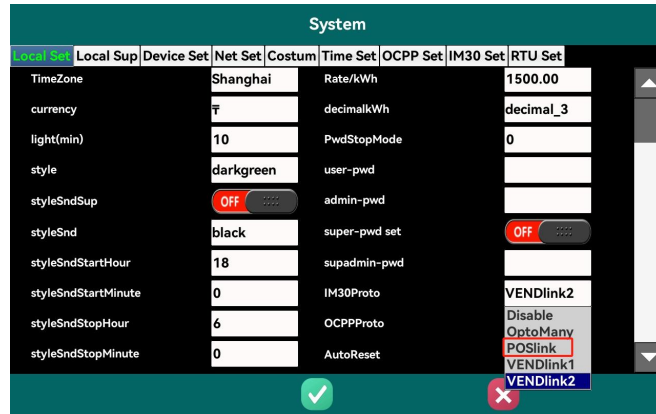
If the authentication is successful, after entering the formal charging process, the charging procedure is as follows: 1.2.4. The charging completion process is as described in 1.2.5 and 1.2.6.

1.5.(Startup Method) Operating Instructions for IM30 Card Reader

The UX pillar is equipped with an IM30 type card reader, which can read credit cards.

Depending on the type of the credit card, users can choose between two card insertion methods, namely the insert type and the slide type.

The start-up method for the POS machine using card swiping requires the background to be opened. The opening path is as shown in the following figure:



In the IM30Proto options, if you select the POSlink mode, the POS machine's card-swiping start button will be enabled on the home page.

1.6.(Startup Method) Password Charging Operation Procedure

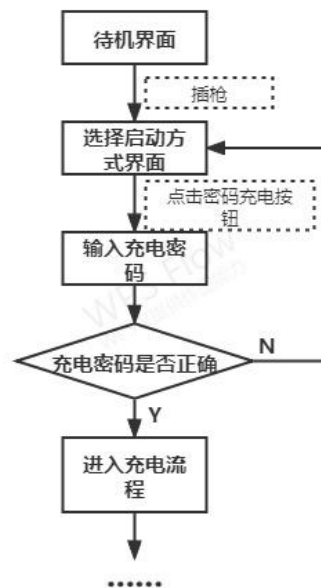
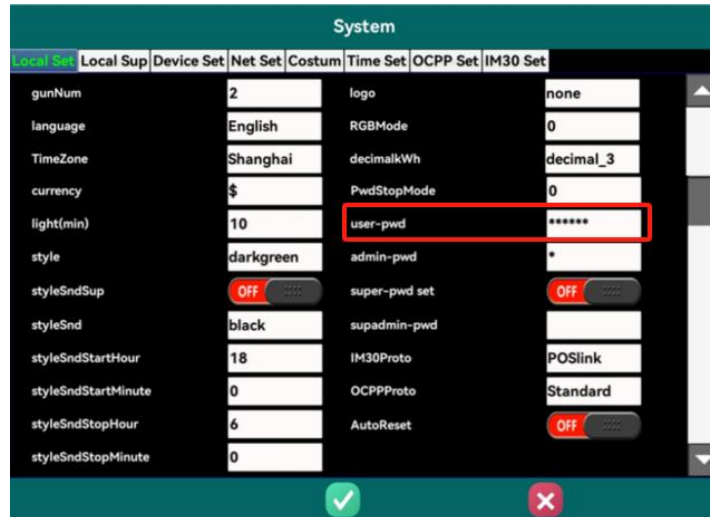


Figure 1-20 Password Charging Process

The password for the password charging startup method needs to be set in the background settings. The opening path is as shown in the following figure:



The current charging password is: 123456

The current end-charging password is: 123456

After the user connects the charging gun to the vehicle, they click the "Password Charging"

button. **PassWord**, The interface will display a digital keypad input box. Once the charging password is correctly entered in this box, the charging startup process can be initiated.

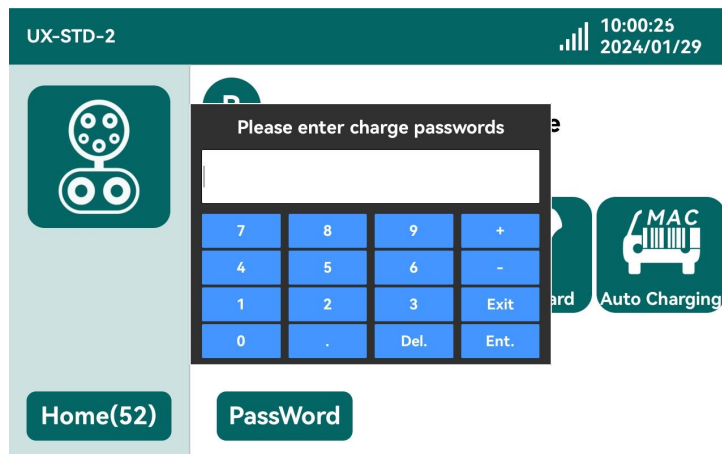


Figure 1-21 Enter the charging password

After entering the correct password, the charging process begins. The subsequent charging procedures can be found in 1.2.4.

1.7.(Startup Method) Automatic MAC Charging

1.7.1.Automatic Charging Operation Procedure for MAC

- 1) The TCU screen background (for the method to enter the background, please refer to 2.1) requires enabling MAC authentication: MAC Auth: "System" --> "Local Sup" --> "MAC

Auth" --> ON;

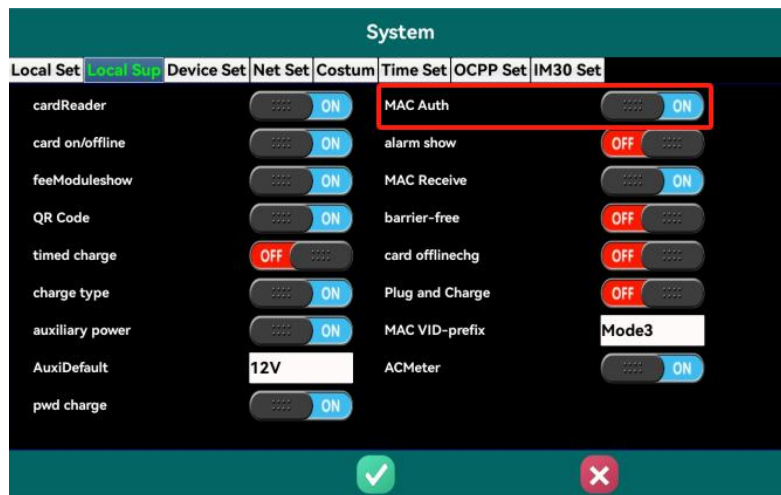


Figure 1-22 MAC Charging Background Support

- 2) After plugging in the charger, select the charging mode as "MAC AutoCharging":

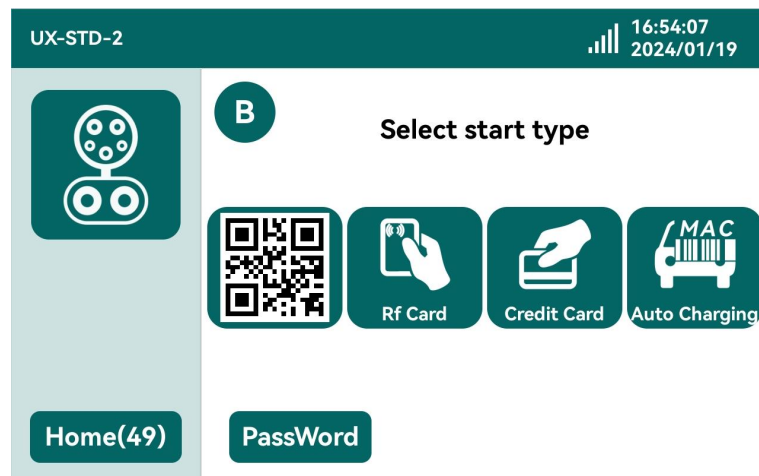


Figure 1-23 MAC Authentication Initiation

- 3) 3) No further actions are required. Charging will commence once the authentication is successful.

1.7.2. Automatic Charging Scheme for MAC

- 1) Background:

In the European standard charging protocol, the vehicle end does not transmit the VIN code; instead, the MAC code is used as a substitute for the VIN code.

- 2) Charging process:

- a. After plugging in the charging gun, TCU will map the MAC address obtained from the vehicle to the VIN code.

- b. After the user selects "MAC AutoCharge" as the startup method, the charging station begins VIN code authentication. It first checks if there is such MAC data in the local database. If not, it proceeds to OCPP authentication. This MAC data serves as the IdTag for the authentication request.
 - c. If the authentication is successful, the authentication data is cleared, and the charging process is initiated. If there is a timeout or failure in the authentication, the authentication data is cleared.
- 3) 3) Local Database: "carinfo.csv"; Vendor: Supplier; Station: Station; Version: Database version number; MAC data consists of 48 bits, which are 12 hexadecimal digits in total. Apart from the MAC data, other data are not yet in use.

	A	B	C	D	E	F	G
1	FileInfo						
2	Vendor	Station	Version				
3	winline	SuZhou-1	1.01				
4							
5	KeyInfo						
6	MAC	VIN	Number-plates	Idtag	parentIdtag	Enable	
7	01FF00000000	dd	dd	dd	dd	dd	
8	141FBA105567	dd	dd	dd	dd	dd	
9	aa-3	dd	dd	dd	dd	dd	
10	aa-4	dd	dd	dd	dd	dd	

Figure 1-24 MAC Local Database

1.8.DCDC/V2G Workflow

- 1) 1) To access the background of the TCU screen (please refer to 2.1 for the method to enter the background), the DDC mode needs to be enabled: System --> Local Settings --> DDC/v2G mode --> Select DDC mode.

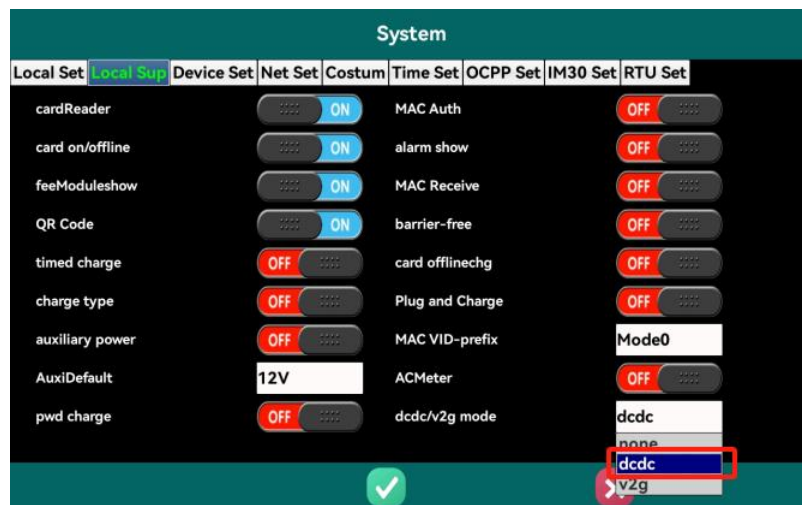


Figure 1-25 DCDC Back-end Support

- 2) After plugging in the charger, select the DC/DC mode.

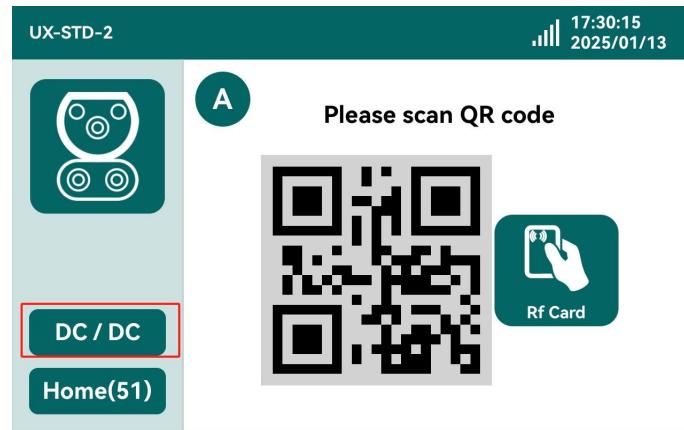


Figure 1-26 Select DCDC mode

- 3) Enter the working mode selection interface and select the DC/DC reverse operation mode: Reserve (discharge).

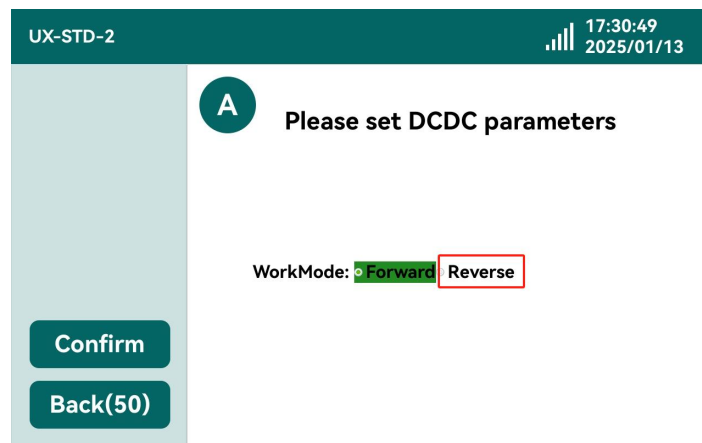


Figure 1-27 Select the ReserveDC/DC reverse mode

- 4) The discharge value can be set in two ways: on the bus side and on the battery side.
5) Select the busbar side and set the specific discharge current value:

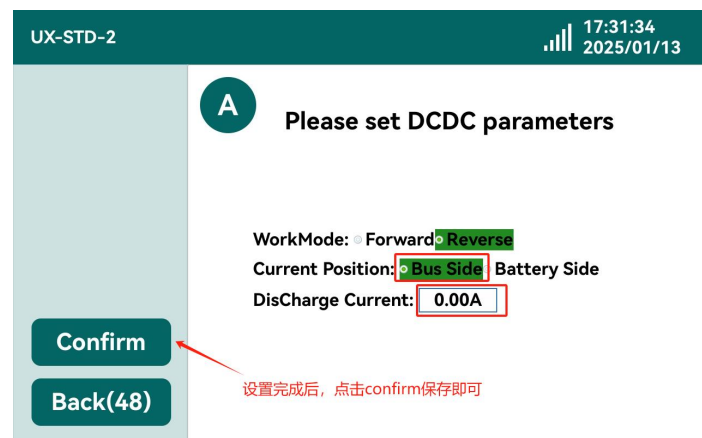


Figure 1-28 Selecting Current Side Data Settings

- 6) Select the battery side, and set the specific discharge voltage and discharge current:

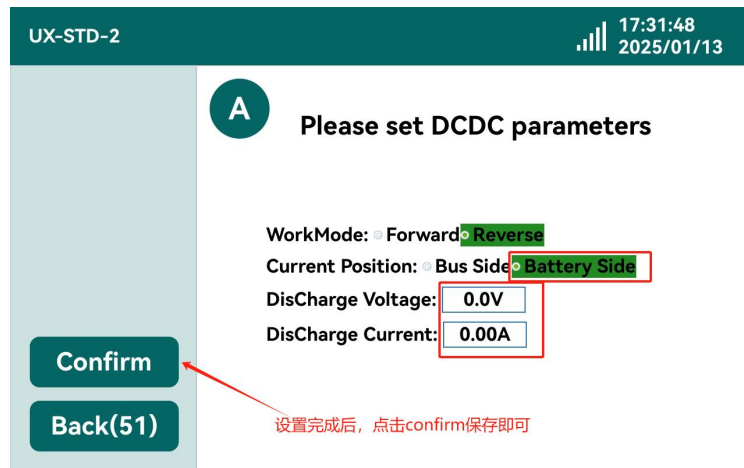


Figure 1-29 Select Battery Side Data Settings

2.Back-end interface

2.1.Enter the background interface

Click the time position on the upper right corner three times continuously. An input keyboard will automatically pop up in the middle of the interface. After correctly entering the password, you can

enter the background settings interface.

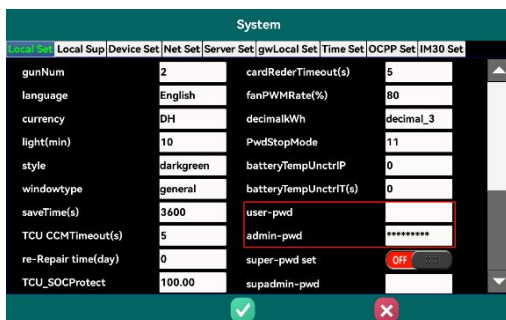
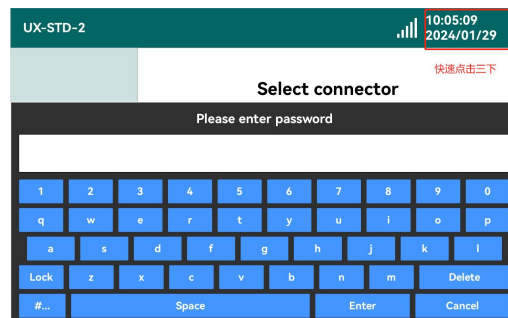


Figure 2-1 Enter the administrator password

Administrator password: The default format for the administrator password is year (last 2 digits) + month + day. For example, if the current date is January 15, 2024, the default password would be: 240115. When the system administrator password (admin-pwd) is configured, the default password becomes invalid and the ID + administrator password should be used.



After the user password (user-pwd) is set, the user viewing function is activated.

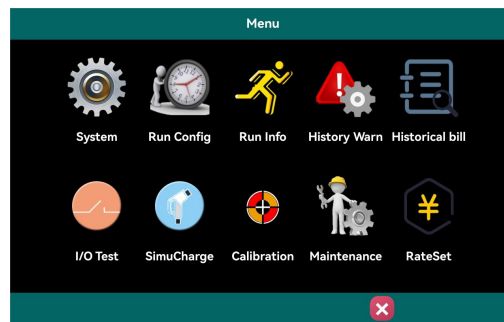


Figure 2-2 Administrator Permission Interface

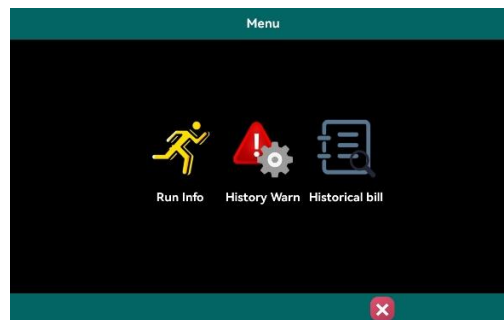


Figure 2-3 User Permission Interface

In the administrator mode, all system and control parameters of the charging station can be set, allowing for hardware strong control debugging, simulation charging, system calibration, equipment maintenance and rate setting.

In the user mode, users can view operation information, fault information and order information.

In the administrator mode, the main menu of the background interface has 10 function buttons corresponding to it: system settings, parameter settings, equipment information, historical faults, historical bills, hardware testing, simulation charging, system calibration, equipment maintenance and rate setting:

- **System Settings:** Configure the charging billing unit (TCU) related settings;
- **Run Configuration:** Set the charging control unit (CCM) related settings;
- **Device Information:** View device communication count, IO feedback, module information, rate and version information;
- **History Warnings:** View local saved fault events;
- **History Bills:** View local saved charging records;
- **Hardware Testing (I/O Test):** Device hardware debugging, forced control of device output ports;

- **Simulated Charging (Simucharge):** Device simulates starting charging and can view detailed charging information;
- **System Calibration (Calibration):** Query and innovate calibration of the coefficient of the mapping function between sample values and actual values;
- **Device Maintenance (Maintenance):** Software upgrade, configuration import/export, log export, and restart control of the charging unit (TCU) and control unit (CCM);
- **Rate Settings (RateSet):** Manually set charging rates.

2.1.1. System Parameter Settings

The managers set the system parameters of the equipment. When the input of the setting values is completed, they need to click the "Confirm" button , Only then can the set parameters be saved. (Note: When it is necessary to switch pages, you need to click the confirm button first, and then proceed with the switch. Otherwise, the previously set parameter values will be restored.)

System							
Local Set	Local Sup	Device Set	Net Set	Costum	Time Set	OCPP Set	IM30 Set
gunNum		2			logo		none
language		English			RGBMode		0
TimeZone		Shanghai			decimalkWh		decimal_3
currency		\$			PwdStopMode		0
light(min)		10			user-pwd		*****
style		darkgreen			admin-pwd		*
styleSndSup		OFF			super-pwd set		OFF
styleSnd		black			supadmin-pwd		
styleSndStartHour		18			IM30Proto		POSlink
styleSndStartMinute		0			OCPPProto		Standard

每页设置完成后，需要点击此处进行保存  

Figure 2-5 System Settings Parameter Interface

2.1.2. Digital Keyboard

When you click on a digital quantity variable, a digital keyboard will pop up.

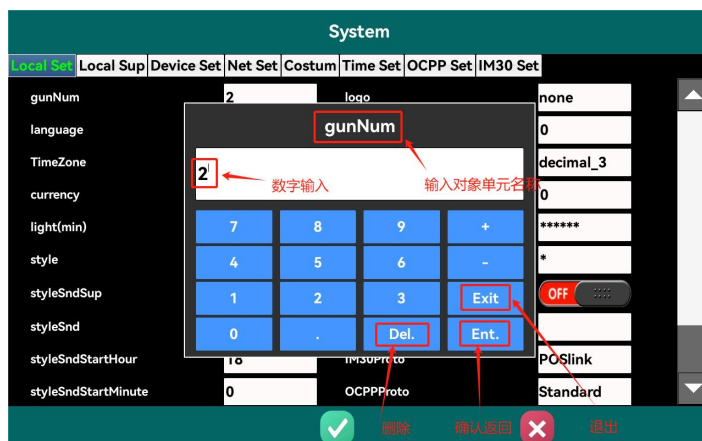


Figure 2-6 Digital Keyboard Interface

2.1.3.Full Keyboard

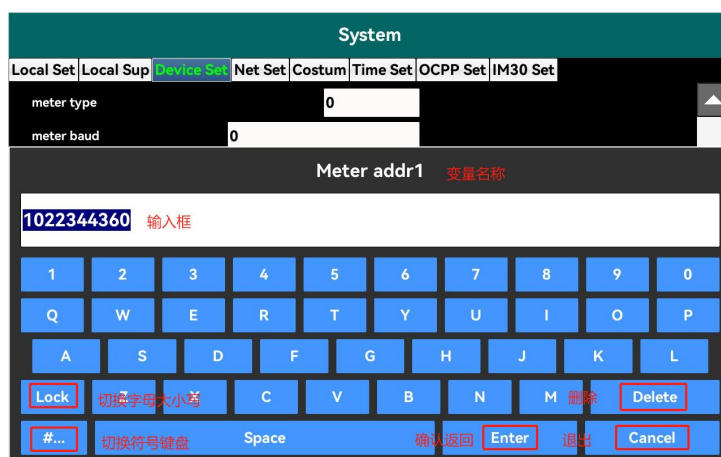


Figure 2-6 Capital Letter Keyboard

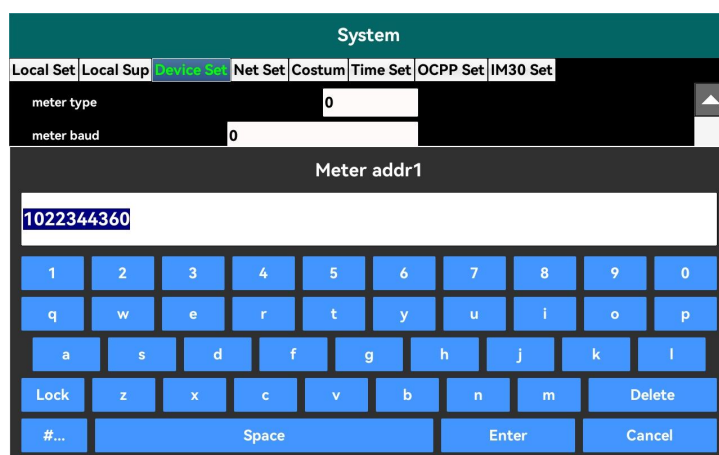


Figure 2-6 Lowercase Letter Keyboard

The image shows a 'System' configuration screen with a tabbed interface. The 'Device Set' tab is selected. It contains fields for 'meter type' (set to 0) and 'meter baud' (set to 0). Below these is a 'Meter addr2' field containing the text '1022344336'. At the bottom is a symbol keyboard with buttons for digits 0-9, punctuation, mathematical symbols, and function keys like 'Delete', 'Enter', and 'Cancel'.

Figure 2-6 Symbol Keyboard

2.2. System Settings

The configuration settings related to the charging station billing unit (TCU) are usually used to configure parameters related to platform communication, network parameters, electricity meter parameters, card reader parameters, etc.

2.2.1. Local Settings

Configuration of local charging station billing unit (TCU) system parameters and charging parameters.

The image shows a 'System' configuration screen with a tabbed interface. The 'Local Set' tab is selected. It displays a list of settings in two columns. The left column includes 'gunNum' (2), 'language' (English), 'TimeZone' (Shanghai), 'currency' (¥), 'light(min)' (1), 'rest light' (20), 'style' (darkgreen), 'styleSndSup' (Off), 'styleSnd' (black), 'styleSndStartHour' (18), 'styleSndStartMinute' (0), 'styleSndStopHour' (6), and 'styleSndStopMinute' (0). The right column includes 'logo' (none), 'RGBMode' (1), 'Rate/kWh' (0.35), 'decimalkWh' (decimal_3), 'PwdStopMode' (0), 'user-pwd' (empty), 'admin-pwd' (empty), 'super-pwd set' (Off), 'supadmin-pwd' (****), 'IM30Proto' (Disable), 'OCPPProto' (Standard), and 'AutoReset' (Off). At the bottom are green and red checkmark/cross buttons.

Figure 2-5 Local Settings Interface

- [gunNum]Number of guns: The total number of guns for the charging stations; dual-guns are set to 2; single-guns are set to 1.
- [language]Language: Interface display language type. Selecting "English" will display the interface in English. The available languages include:

English	英语
German	德语
Polish	波兰语
Chinese	简体中文
Spanish	西班牙语
Arabic	阿拉伯语
Traditional	繁体中文
Vietnamese	越南语
Magyar	匈牙利语
Indonesian	印度尼西亚语
Italian	意大利语
Malay	马来语
Thai	泰语
French	法语
Portugues	葡萄牙语
Russian	俄语
Armenian	亚美尼亚语

c. [TimeZone]Time Zone: Time zone selection. Choose the time zone of your actual location.

The available time zones include:

Asia	Shanghai	上海	中国
	Bangkok	曼谷	泰国
	Dubai	迪拜	阿拉伯
	Manila	马尼拉	菲律宾
	Singapore	新加坡	新加坡
	Almaty	阿拉木图	哈萨克斯坦
	Tehran	德黑兰	伊朗
	Katmandu	加德满都	尼泊尔
Europe	London	伦敦	英国
	Malta	马耳他	马耳他

	Stockholm	斯德哥尔摩	瑞典
	Madrid	马德里	西班牙
	Berlin	柏林	德国
	Dublin	都柏林	爱尔兰
	Warsaw	华沙	波兰
	Sofia	索菲亚	保加利亚
	Istanbul	伊斯坦布尔	土耳其
	Helsinki	赫尔辛基	芬兰
	Lisbon	里斯本	葡萄牙
Australia	Sydney	悉尼	澳大利亚
Atlantic	Reykjavik	雷克雅未克	冰岛
Africa	Cairo	开罗	埃及
	Kigali	基加利	卢旺达
America	Argentina	圣保罗	阿根廷、巴西、乌拉圭
America	Lima	利马	秘鲁
Pacific	Tahiti	大溪地	法属波利尼西亚

d. [Rate/kWh]Rate: Electricity charge per kilowatt-hour (kWh);

e. [currency]currency:

¥	人民币	中国
€	欧元	马耳他、西班牙、德国、爱尔兰、芬兰、葡萄牙
£	英镑	英国
\$	美元	美国
kr	瑞典克朗	瑞典
zł	兹罗提	波兰
₺	里拉	土耳其
S\$	新加坡元	新加坡
лв	列弗	保加利亚
₽	卢布	俄罗斯
A\$	澳元	澳大利亚

DH	迪拉姆	阿拉伯
RM	林吉特	马来西亚
₪	谢克尔	以色列
T	哈萨克斯坦坚戈	哈萨克斯坦
đ	越南盾	越南
B	泰铢	泰国
﷼	伊朗里亚尔	伊朗
Br	埃塞俄比亚比尔	埃塞俄比亚
₱	菲律宾比索	菲律宾
₮	蒙古图格里克	蒙古
Lei	罗马尼亚列伊	罗马尼亚
Ft	福林	匈牙利
Rp	卢比	印度尼西亚
\$U	乌拉圭比索	乌拉圭
S/.	秘鲁新索尔	秘鲁
	不显示货币符号	不显示货币符号
Rs	斯里兰卡卢比	斯里兰卡
₣	法郎	法属波利尼西亚
R\$	巴西雷亚尔	巴西
£	埃及镑	埃及
HK\$	港币	中国香港
RF	卢旺达法郎	卢旺达
֏	亚美尼亚德拉姆	亚美尼亚

- f. [light] Screen-off time (min): The time it takes for the interface to enter the screen-saver state when there is no operation for a continuous period;
- g. [rest light] Screen-off brightness (%): The brightness of the interface when entering the screen-saver state after no operation for a continuous period.
- h. [Style] Theme color: black; darkgreen; seablue; orange;
- i. Lightstyle: Bright style; DarkStyle: Dark style.

- j. [StyleSndSup] Main interface skin switching function;
- k. [StyleSnd] Second style switching function (different from the default style); black; darkgreen; seablue; orange; Lightstyle; DarkStyle.
- l. [StyleStartHour] Starting time of the second style - hour;
- m. [StyleStartMinute] Starting time of the second style - minute;
- n. [StyleStopHour] Ending time of the second style - hour;
- o. [StyleStopMinute] Ending time of the second style - minute;
- p. [logo] Display of the main interface logo; 0 - disabled; 1 - Yonglian; 2 - Australian EVUP; 3 - Turkish CW; 4 - chargesini; 5 - Dahua; 6 - MSI;
- q. [RGB Mode] Display on the lamp panel: The setting for dual guns and single light bar is 0.
- r. [DecimalkWh] Decimal places for charging capacity; decimal-3 is 3 digits; decimal-2 is 2 digits.
- s. [PwdStopMode] Charging stop mode: The way to stop charging when charging is in progress: 0 - directly stop charging; non-zero: When swiping the card starts the process, the card swipe stop function needs to be used. To test the start, the password "123456" needs to be entered; for other start methods, the password "1234" needs to be entered;
- t. [usr-pwd] User password: Password for starting charging. When this password is empty, the default password for starting charging is 123456; the password for entering the user (permission) background menu page.
- u. [admin-pwd] Administrator Password: The password for accessing the administrator (privileged) backend menu page. When this configuration is empty, the default format of the administrator password is: year (last 2 digits) + month + day. For example, if the current date is January 15, 2024, the default password would be: 240115. When the system administrator password (admin-pwd) is configured, the default password becomes invalid and is replaced with the station ID + administrator password.
- v. [Super-pwd set] Super Password Setting Switch; "not support" indicates not supported, "support" indicates supported.
- w. [supadmin-pwd] Super Password: The password for accessing the super administrator (authority) backend menu page. The default format of the super administrator password is: year (last 2 digits) + month + day. For example, if the current date is January 23, 2024, the

default password would be: 240123. When the super password can be configured and enabled, and the currently set super password is not empty, use the current configured super password to enter the super administrator backend. The default password becomes invalid.

x. [IM30Proto] IM30 POS Machine Protocol:

- Disable -- Disable: In this state, the M1 card is used for card-swiping to initiate charging.
- OptoMany:
- POSlink: Includes both credit card and M1 card modes.
- VENDlink1: In this state, the preauth (freeze first, then deduction) mode is adopted.
- VENDlink2: In this state, the sale/refund (deduction first, and the excess money is refunded later) mode is adopted.

y. [OCPPProto] ocpp protocol type:

- N/A: Reserved
- Standard: OCPP standard
- Custom1: OCPP Start Transaction message, including Start SOC
- Custom2 - Custom10: Customized protocols

z. [AutoReset] OCPP network exception handling: Whether to enable offline reset function.

2.2.2.Local Support

The system functions supported by the charging station billing unit (TCU) system.

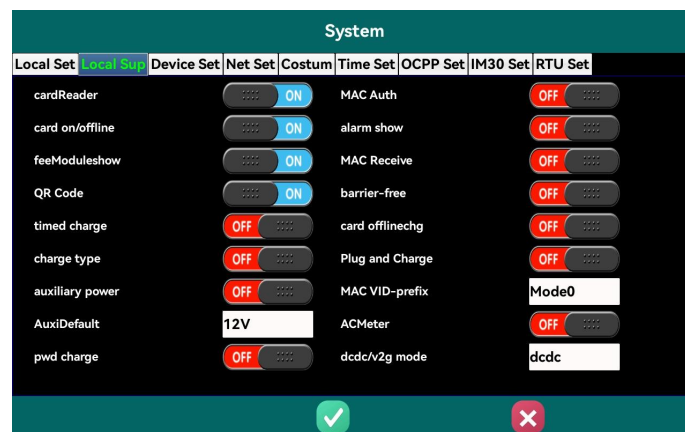


Figure 2-7 Local Support Interface

a. [cardReader]Card Reader Support: M1 card startup charging support. If credit card support is turned off simultaneously, the M1 card reader will be enabled by default, meaning this option is invalid at this time;

- b. [card on/offline]Offline Card/Online Card: M1 card type selection. Turning off is for offline card, and turning on is for online card. When starting with an M1 card, offline cards do not require authentication, while online cards need to authenticate with the platform;
- c. [feeModuleshow]Fee Model Display Support: Charging interface displays rates and charging amounts. When this function is turned off, all display related to amount information on the charging page will be hidden.
- d. [QR Code]QR Code; Whether to allow scanning to start charging. Refer to 1.2.1 Scanning to Start Charging Selection.
- e. [timed charge]Timed Charging Support: Timed startup charging function support;
- f. [charge type]Charge Type Selection Support: Refer to 1.2.2 Selection of Four Charging Modes;
- g. [auxiliary power]Auxiliary Power Selection Support: Auxiliary power switching function support; European standard without auxiliary power.
- h. [AuxiDefault]Default Auxiliary Power: Default auxiliary power voltage (12V/24V);
- i. [pwdcharge>Password Charging Support: Password startup charging function support;
- j. [MAC Auth]MAC Authentication Support: European standard charging protocol does not upload VIN code, using MAC instead of VIN code;
- k. [alarm show]Alarm Display: Charging in progress alarm display function support;
- l. [MAC Receive]When using password charging or cardless charging, MAC is used to fill UsrID;
- m. [barrier-free]Customer Customization Requirements;
- n. [card offlinechg]M1 Offline Card Swipe Charging Function Support: After enabling this function, M1 card startup does not require authentication when the network is disconnected;
- o. [MAC VID-prefix] The EU standard MAC code replaces the VIN code. The selection mode is: model 0/1/2/3;
 - Mode 0: 12-bit MAC original characters (112233445566)
 - Mode 1: MAC with VID prefix added (VID112233445566)
 - Mode 2: Separated by colon (11:22:33:44:55:66)
 - Mode 3: Separated by VID + colon (VID:112233445566)
- p. [Plug and Charge]Support for plug-and-charge.

q. [dcdc/v2g mode]Whether to select dcdc/v2g mode.

- none: Normal charging mode
- dcdc: dcdc mode.

2.2.3.Equipment Settings

Settings for the product parameters of charging piles, the parameters of electricity meters, and the parameters of card readers.

System								
Local Set	Local Sup	Device Set	Net Set	Customer	Time Set	OCPP Set	IM30 Set	RTU Set
meter type		0						
meter baud		0						
meter splitter		1.00						
meter accuracy		1000.00						
Meter addr1								
Meter addr2								
meter model								
meter Soft version								
KEM addr1		0						
KEM addr2		0						

System								
Local Set	Local Sup	Device Set	Net Set	Customer	Time Set	OCPP Set	IM30 Set	RTU Set
KEM addr2		0						
KEM CT0		200						
KEM CT1		5						
KAVASS addrA		00000000000000						
KAVASS addrB		00000000000000						
CardArea		2						
CardType		default						
cardAnalysis		Packed BCD						
cardReadBytes		0						
cardStartBytes		0						

System								
Local Set	Local Sup	Device Set	Net Set	Customer	Time Set	OCPP Set	IM30 Set	RTU Set
CardType		default						
cardAnalysis		Packed BCD						
cardReadBytes		0						
cardStartBytes		0						
1CardPwd		64						
2CardPwd		226						
3CardPwd		1						
4CardPwd		0						
5CardPwd		74						
6CardPwd		60						

Figure 2-9 Device Settings Interface

- a. [meter type]Electric meter type (not in use yet): Electric meter type selection; (default) 0 - standard, 1 - (not available yet).
- b. [meter baud]Electric meter baud rate (not in use yet): Electric meter baud rate setting; DCM (Dong Hong) electric meter is set to 2400.
- c. [meter fenliuqi]Electric meter multiplier: Related to the shunt resistor coefficient (DC current), the coefficient is set to 1 by default.
- d. [meter accuracy]Electric meter accuracy: Electric meter accuracy setting (for different brands of electric meters, such as Yada dual-channel electric meters, Dong Hong, Schneider electric meter communication protocol accuracy is 0.001, set to 1000; 645 electric meter's protocol accuracy is 0.01, set to 100); (unit: Wh)
- e. [meter addr1]Electric energy meter address 1: Represents the address of one channel for multi-channel electric meters; (read/write according to the number on the electric meter) applicable to Yada dual-channel DC meters
- f. [meter addr2]Electric energy meter address 2: Represents the address of the second channel for multi-channel electric meters; (read/write according to the number on the electric meter) applicable to Yada dual-channel DC meters
- g. [meter model]Electric meter type
- h. [meter Soft version]Electric meter software version number
- i. [KEM addr1]Electric energy meter Modbus address 1: Modbus address, default is 001. (range 001-255)
- j. [KEM addr2]Electric energy meter Modbus address 2: Modbus address, default is 002. (This needs to be set for the second electric meter in the case of dual guns) (range 001-255)
- k. [KEM CT0]Electric energy meter CT value: Electric meter CT value setting, set according to the protocol of different electric meters.
- l. [KEM CT1]Electric energy meter CT value: Electric meter CT value setting, set according to the protocol of different electric meters. (This needs to be set for the second electric meter in the case of dual guns)
- m. [KAVASS addrA]Electric energy meter ground lock address 1;
- n. [KAVASS addrB]Electric energy meter ground lock address 2;

- o. [CardArea]Card sector: Yonglian card sector is set to 2;
- p. [CardType]Card type: Yonglian card serial number is 0, the card reader can recognize the charging card types (see Table 1): 0 Yonglian card; 1 Yunkuai card; 2 Singapore elc; 3 Turkey cw; 4 Vega card
- q. [cardAnalysis]Card number parsing mode: 0 - compressed BCD; 1 - ASCII; 2 - BCD code
- r. [cardReadBytes]Card number parsing byte length
- s. [cardStartBytes]Card number parsing starting byte
- t. [1CardPwd]-[6CardPwd]: Card key

2.2.4.Network Settings

Network parameter settings and viewing of the charging station billing unit (TCU).

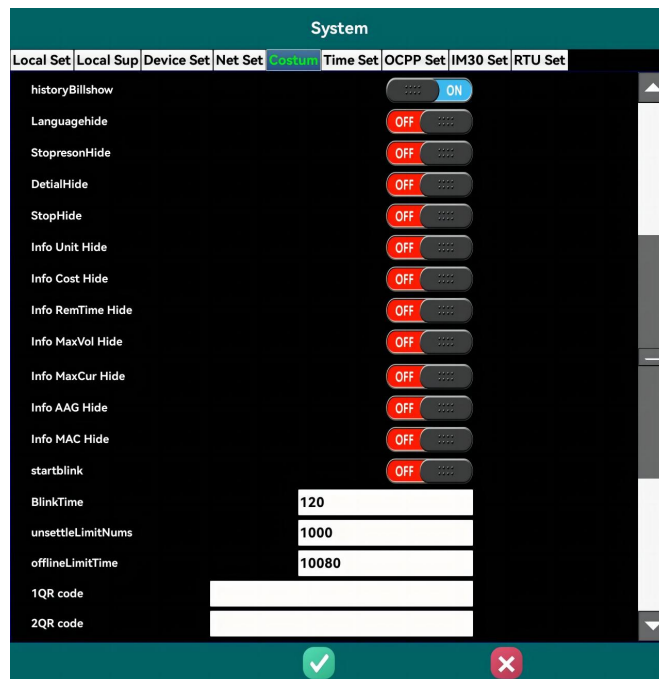
System								
Local Set	Local Sup	Device Set	Net Set	Costum	Time Set	OCPP Set	IM30 Set	RTU Set
RouterType			seriallink					
NetType			1					
IPAddr1			192.168.8.99					
Mask1			255.255.255.0					
Gway1			192.168.8.1					
Dns1			8.8.8.8					
Current IP			192.168.8.99					
Current Mask			255.255.255.0					
Current MAC			66:6E:5B:A6:72:42					
Router username			admin					
Router password			*****					

Figure 2-12 Network Settings Interface

- a. [Router Type]Router Type Selection: Disable, seriallink, hera-604 and hongdian (Fuhdi), please select according to the specific model of the router you are using. For UX jian, select seriallink.
- b. [Net Type]Wired Network Type: 0 - Dynamic IP acquisition, 1 - Static IP
- c. [IPAddr1]IP Address 1: IP address setting for network interface 1 (eth0) (only valid for Net Type = 1);
- d. [Mask1]Subnet Mask 1: Subnet mask setting for network interface 1 (eth0) (only valid for Net Type = 1);
- e. [Gway1]Gateway 1: Gateway setting for network interface 1 (eth0) (only valid for Net Type = 1);
- f. [Dns1]Domain Name System 1: DNS setting for network interface 1 (eth0);

- g. [Current IP]Current IP Address: IP address setting for current network interface (eth0);
- h. [Current Mask]Current Subnet Mask: Subnet mask setting for current network interface (eth0);
- i. [Current MAC]Current Gateway: Gateway setting for current network interface (eth0);
- j. [Router username]Router Username: View router username;
- k. [Router password]Router Login Password: View router login password.

2.2.5.OCPP Protocol Type Function Settings



- a. [historyBillshow]: The function key for displaying historical bills is supported;
- b. [languagehide]: The language switch on the main interface is hidden;
- c. [StopreasonHide]: The stop reason is hidden;
- d. [DetailHide]: The detail button on the charging interface is hidden;
- e. [StopHide]: Whether the stop button on the charging interface is displayed;
- f. [Info Unit Hide]: The unit price is hidden;
- g. [Info Cost Hide]: The charging amount is hidden;
- h. [Info RealTime Hide]: The charging time is hidden;
- i. [Info MaxVol Hide]: The maximum charging voltage is hidden;
- j. [Info MaxCur Hide]: The maximum charging current is hidden;
- k. [Info AAG Hide]: The signal display is hidden;

- l. [Info MAC Hide]: The MAC display is hidden;
- m. [startblink]: Remote start blink (customized);
- n. [BlinkTime]: Interval time for remote start blink;
- o. [unsettleLimitNums]: Limit number of unsettled orders (customized);
- p. [offlineLimitTime]: Time limit for offline operation (customized);
- q. [1QR code]: Charging QR code for gun A;
- r. [2QR code]: Charging QR code for gun B.

2.2.6.Time Settings

System								
Local Set	Local Sup	Device Set	Net Set	Costum	Time Set	OCPP Set	IM30 Set	RTU Set
year			2024					
month			12					
day			2					
hour			20					
min			58					
sec			42					
Lightting-off(hour)			8					
Lightting-off(min)			0					
Lightting-on(hour)			16					
Lightting-on(min)			0					

Figure 2-17 Time Setting Interface

a. Parameter Correspondence Table:

Year	年	Month	月
Day	日	Hour	小时
Min	分	Sec	秒
Lightting-off(hour)		每日照明灯关闭时间(时)	
Lightting-off(min)		每日照明灯关闭时间(分)	
Lightting-on(hour)		每日照明灯开启时间(时)	
Lightting-on(min)		每日照明灯开启时间(分)	

2.2.7.OCPP Platform Configuration

The screenshot displays the 'System' configuration page with the 'OCPP Set' tab selected. The interface includes a top navigation bar with tabs: Local Set, Local Sup, Device Set, Net Set, Costum, Time Set, OCPP Set, IM30 Set, and RTU Set. The main area contains the following settings:

- ICCID: (empty field)
- IMSI: (empty field)
- OCPPId: UX-STD-2
- Device Id: CED888888888
- Server url: ws://192.168.161.23:8080/steve/webs
- Auth1 Mode: Disable
- Auth1 username: NA
- Auth1 password: NA
- SwDoubleOCPP: Disable
- OcppVice Id: (empty field)
- SerVice url: (empty field)
- Auth2 Mode: Disable
- Auth2 username: NA
- Auth2 password: NA
- AllowOfflineTxForUnknownId: ON
- AuthorizationCacheEnabled: OFF
- LocalAuthorizeOffline: ON
- LocalPreAuthorize: OFF
- StopTransactionOnInvalidId: ON
- LocalAuthListEnabled: ON
- HeartbeatInterval: 60
- WebSocketPingInterval: 50
- chargemodel: UXD-160
- chargervendor: winline

At the bottom, there are green and red checkmark icons.

Figure 2-18 OCPP Settings Interface

- [ICCID] SIM card's ICCID: Used to view the SIM ICCID information when the 4G card is connected to the network;
- [IMSI] SIM card's IMSI: Used to view the SIM IMSI information when the 4G card is connected to the network;
- [OCPPId]: The ID used for OCPP login (default is the main platform);
- [Device Id]:桩序序列号;
- [Server url]: The domain name for OCPP connection;
- [Auth1 Mode]: HTTP authentication mode;
- [Auth1 username]: HTTP authentication username;
- [Auth1 password]: HTTP authentication password (default is the main platform);
- [SwDoubleOCPP]: Whether to enable the secondary platform , for dual-platform connection;

- j. [OcppVice Id]: The OCPPID of the secondary platform;
- k. [SerVice url]: The domain name for the secondary platform connection;
- l. [Auth2 Mode]: The HTTP authentication mode of the secondary platform;
- m. [Auth2 username]: The HTTP username of the secondary platform;
- n. [Auth2 password]: The HTTP authentication password of the secondary platform;
- o. [AllowOfflineTxForUnknownId]: Automatically authenticate unknown accounts in offline status;
- p. [AuthorizationCacheEnabled]: Whether to allow using Cache for local authentication;
- q. [LocalAuthorizeOffline] : Whether to support local authentication during offline (Cache/List);
- r. [LocalPreAuthorize]: Whether to support local authentication (Cache/List);
- s. [StopTransactionOnInvalidId]: When restoring online status after offline, if the charging station is still in the charging state and the StartTransaction.conf status is not Accepted:
 - At this point, if it is set to true: Continue charging;
 - if it is set to false: Stop charging;
- t. [HeartbeatInterval]: Heartbeat message sending period (unit: S);
- u. [WebSocketPingInterval]: Period for checking if WebSocket is online (unit: S);
- v. [chargermodel]: Charging station model;
- w. [chargervendor]: Charging station supplier;

2.2.8.IM30 Card Reader Settings

UX guideline: The IM30 card reader communicates via serial port. This interface option does not require any configuration.

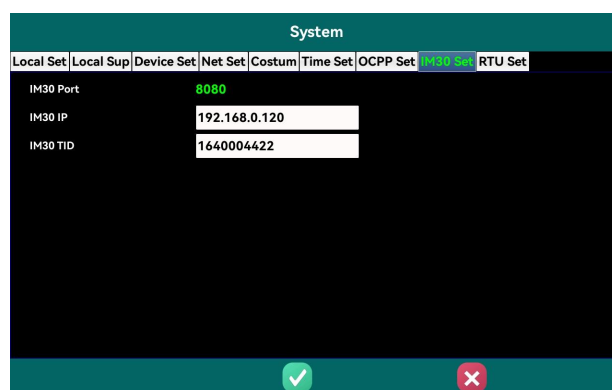


Figure 2-19 IM30 Card Reader Settings Interface

- a. [IM30 Port]: The network port number used by the card reader.
- b. [IM30 IP]: The IP address of the IM30 card reader.
- c. [IM30 TID]: The serial number code on the back of the card reader.

2.2.9.RTU Settings

System	
Local Set	Local Sup
Device Set	Net Set
Costum	Time Set
OCPP Set	IM30 Set
RTU Set	
RTU Port	502
RTU IP	192.168.8.99

- a. [RTU Port]: Modbus port number.
- b. [RTU IP]: IP address of the Modbus server.

2.3.Parameter Settings

The configuration settings for the main control unit (CCM) of the charging station are used to configure the support for fault detection of the charging station, fault detection thresholds, parameters related to the charging module, alarm levels, etc. (Note: The parameter settings for CCM require communication through the 485 serial port of the gsp backend. Therefore, the gsp backend communication ports of the TCU and the CCM need to be connected, and then in the parameter settings -> charging settings, the backend communication port should be configured as the CCM communication; the current version only supports a single CCM structure and cannot mount multiple terminals.)

2.3.1.Function Sup



Figure 2-17 Function Support Interface

Settings for the charging-related functions and fault detection functions supported by the main control unit of the charging station.

- a. CheckK1K2OutVol: Detection of the external voltage of K1K2, the detection function for the external voltage of DC contactors K1K2 being greater than 10V is supported;
- b. CCM_Door: CCM - System access control, the access control alarm function of the main control unit (CCM) is supported (if the access control feedback is received by CCM, this function needs to be enabled in order to detect access control faults);
- c. PCM_Door: PCM - System access control, the access control alarm function of the power control unit (PCM) is supported (if the access control feedback is received by PCM, this function needs to be enabled in order to detect access control faults);
- d. BMS_BatteryErr: BMS - Single battery fault detection, the fault alarm functions for over-temperature and over-voltage of a single battery of BMS are supported;
- e. GunTempErr: Over-temperature alarm for the charging gun is supported, the alarm function for over-temperature of the charging gun temperature is supported;
- f. K3K4Cut: Support for rapid cut-off of K3K4, when a fault occurs, the auxiliary power switch K3K4 is directly disconnected, the function support for this is provided;
- g. EleLockErr: Support for electronic lock faults, the fault detection function for the electronic lock of the charging gun is supported;
- h. BSM_ErrStopChg: Support for terminating charging due to BSM faults, the function support

for terminating charging when BSM faults occur is provided;

- i. CheckGunInHolder: Support for gun head insertion detection, the detection function for the insertion of the charging gun head is supported;
- j. BCP DcSamp: Support for maximum voltage sampling of BCP, the function support for maximum voltage sampling of BCP is provided (default support);
- k. PreAlarmNotLight: Alarm light for insulation warning not on, the function support for the alarm light for insulation warning not being on is provided;
- l. CheckK1K2K1K2: Feedback detection support, the detection function for the feedback of DC contactors K1K2 is supported;
- m. CheckCurOver: Overcurrent detection support, the detection function for overcurrent of the charging output current is supported;
- n. CCM_NoValidCur: Support for no valid current in CCM, the fault detection function for no valid current in the output of the main control unit (CCM) is supported;
- o. EleLockNegtiveLogic: Support for negative logic of the electronic lock, the function support for negative logic of the electronic lock is provided;
- p. SecondUnlock: Support for secondary unlocking, the function support for the electronic lock control for secondary unlocking of the charging gun is provided;
- q. EleLockCloseCtrl: Support for the closing of the gun electronic lock, the function support for the control of closing the electronic lock of the charging gun is provided;
- r. CCM_BothCharge: Support for parallel charging in CCM, the charging function of the main control unit (CCM) is supported;
- s. BothStepCharge: Support for parallel charging across poles, the function support for parallel charging across poles is provided;
- t. CheckAcSwitchErr: Support for fault detection of the AC contactor, the function support for fault detection of the AC contactor is provided;
- u. CheckFanErr: Support for system fan fault detection, the function support for system fan fault detection is provided;
- v. DYCPowePatch: Support for dynamic power allocation, the dynamic scheduling function of the charging module group is provided;
- w. OneBusCircleMode: Support for the single bus round charging scheme of the system, the

system single bus round charging scheme (multiple gun charging stations, and only one bus, this electrical scheme does not support simultaneous charging of multiple guns, only the round charging scheme can be adopted) is provided;

- x. CheckAcInput: Support for AC input fault detection, the function support for AC input fault detection is provided;
- y. EqualCharaeMode: Power equal distribution, when the system has more than one gun with a demand greater than the actual, equal distribution is required (the small orange flexible distribution needs to support it);
- z. singleRmErrNotStop: Support for not stopping the machine in case of a single module fault: In the process of charging, a single module fault does not stop the charging function is supported; PreChgPlanB: Precharge Plan B support, Precharge B plan support (0 - Old A plan; 1 - New B plan);
- aa. PreChg20ov: Precharge 200V support, 200V through support for Precharge B plan;
- ab. K1K2PreClose: Authentication pre-closure of K1K2, in authentication mode, the module starts with K1K2 closed strategy supported.

2.3.2.Electrical Settings

The settings for charging-related parameters and fault thresholds of the main control unit (CCM) of the charging station can be simulated to trigger fault events on this page.

Run Config					
Function Sup	Charge Set	Module Set	Alarm Set	Chg Manage	Other Set
gun1 type	CCS Combo		PreChrgVolOffset	5.00	
gun2 type	CCS1 Combi		GunCntLedTwinkle	0	
singleNo	Disable		CCM_SOCProtect	99	
Gun1MaxVol	1000.00		PreChrgMaxVol	1000.00	
Gun1MaxCur	200.00		PreChrgMinVol	105.00	
Gun1MinVol	105.00		PreChrgWaitTimer	3	
Gun1MinCur	1.00		Gun1PowerLimit	160.00	
Gun2MaxVol	1000.00		Gun2PowerLimit	160.00	
Gun2MaxCur	200.00		SysPowerLimit_TCU	1000.00	
Gun2MinVol	105.00		OverCurTimer(ms)	1000	
Gun2MinCur	1.00		CCM_SerialComm	ccm_485 coi	
GunTempErrValue	90.00		IsoAlarmLimit	500.00	
GunTempWarnValue	80.00		IsoErrLimit	100.00	
GunTempWarnBack	5.00		preKwh1(kWh)	0	
GunTempPwrLimit	0.60		preSoc1(%)	0	
GunResmOumu	0.00		preMin1(min)	0	
DYCPatchTimer	30		TimeChgHour	0	
IOSCheckTimer	4		TimeChgMin	0	
BMSCurReqRatio	1.00		A_proto	0	
SelfCheckVolOffset	5.00		B_proto	0	

Figure 2-19 Charging Settings Interface

- a. gun1 type: Charging gun protocol standard; CCS Combo European standard.
- b. gun2 type: Charging gun protocol standard; CHADEMO Japanese standard; CCS...
MBO American standard; GBT Chinese standard
- c. singleNo: Single gun pole configuration; Disable - Disable; A - Only gun A; B - Only gun B
- d. Gun1MaxVol: Maximum voltage of gun 1 charging gun; Maximum output voltage (limit) value of the charging gun; (Unit: V)
- e. Gun1MaxCur: Maximum current of gun 1 charging gun; Maximum output current (limit) value of the charging gun; (Unit: A)
- f. Gun1MinVol: Minimum voltage of gun 1 charging gun; Minimum output voltage (limit) value of the charging gun; (Unit: V)
- g. Gun1MinCur: Minimum current of gun 1 charging gun; Minimum output current (limit) value of the charging gun; (Unit: A)
- h. Gun2MaxVol: Maximum voltage of gun 2 charging gun; Maximum output voltage (limit) value of the charging gun; (Unit: V)

- i. **Gun2MaxCur:** Maximum current of gun 2 charging gun; Maximum output current (limit) value of the charging gun; (Unit: A)
- j. **Gun2MinVol:** Minimum voltage of gun 2 charging gun; Minimum output voltage (limit) value of the charging gun; (Unit: V)
- k. **Gun2MinCur:** Minimum current of gun 2 charging gun; Minimum output current (limit) value of the charging gun; (Unit: A)
- l. **GunTempErrValue:** Over-temperature fault value of the charging gun;
Over-temperature warning threshold of the charging gun (power reduction point, only warning prompt, no power-off charging); (Unit: °C)
- m. **GunTempWarnValue:** Over-temperature warning value of the charging gun;
Over-temperature warning threshold of the charging gun (power reduction point, only warning prompt, no power-off charging); (Unit: °C)
- n. **GunTempWarnBack:** Over-temperature warning offset value of the charging gun;
Over-temperature warning threshold offset value of the charging gun (power reduction point); (Unit: °C)
- o. **GunTempPwrLimit:** Over-temperature power reduction percentage of the charging gun;
Over-temperature limit power percentage of the charging gun; Maximum power over-temperature = Maximum output power × Over-temperature power reduction percentage%; (Unit: %)
- p. **DYCPatchTimer:** Dynamic power allocation interval, Dynamic request power allocation time interval; (Unit: s)
- q. **IOSCheckTimer:** Insulation detection maximum time, Insulation detection maximum waiting time; (Unit: s)
- r. **BMSCurReqRatio :** BMS demand current coefficient; BMS demand current coefficient (for some models, it is prone to trigger BSM overcurrent, multiplied by the coefficient to be normal);
- s. **SelfCheckVolOffset:** Self-check voltage offset; Self-check voltage offset value (self-check voltage can be set, to solve the problem of module startup timeout); (Unit: V)
- t. **PreChrgVolOffset:** Pre-charge voltage offset; Pre-charge voltage offset value (pre-charge voltage offset can be set, to solve the problem of module pre-charge failure);

(Unit: V)

- u. **GunCntLedTwinkle:** LED that flashes when the gun is plugged in, flashing function of the LED strip for the charging gun and vehicle connection (when the setting value is 1, this function is enabled);
- v. **CCM_SOCProtect :** CCM-SOC charging protection value, Maximum SOC protection value of the main control unit (CCM); (Unit: %)
- w. **PreChrgMaxVol:** Maximum pre-charge voltage; Maximum charging voltage during pre-charge; (Unit: V)
- x. **PreChrgMinVol:** Minimum pre-charge voltage; Minimum charging voltage during pre-charge; (Unit: V)
- y. **PreChrgWaitTimer:** Pre-charge waiting delay time: Pre-charge waiting delay time; (Unit: s)
- z. **Gun1PowerLimit:** Power limit of gun 1; Limit power value of gun 1; (Unit: kW)
- aa. **Gun2PowerLimit:** Power limit for gun 2, the maximum power value for the 2nd charging gun; (Unit: kW)
- ab. **SysPowerLimit_TCU:** Maximum charging power limit through TCU.
- ac. **BatOverTemp:** Over-temperature protection value for the battery, the temperature over-temperature protection value for the battery; (Unit: ° C)
- ad. **OverCurTimer(ms):** Overcurrent filtering time (ms), the filtering time for overcurrent faults of the charging gun;
- ae. **CCM_SerialComm:** Backend communication port configuration, gsp backend communication object setting (CCM communication: communication with the main control unit (CCM), gsp backend communication: communication with the computer upper computer);
- af. **Gun1FaultIndex:** Fault index for gun 1, the simulated fault code for gun 1;
- ag. **Gun1FaultTrig:** Fault trigger for gun 1, the simulated fault trigger switch (set to open will trigger the fault set by the gun 1 fault index, and needs to click the confirmation button to take effect);
- ah. **Gun2FaultIndex:** Fault index for gun 2, the simulated fault code for gun 2;
- ai. **Gun2FaultTrig:** Fault trigger for gun 2, the simulated fault trigger switch (set to open

will trigger the fault set by the gun 1 fault index, and needs to click the confirmation button to take effect);

- aj. **IsoAlarmLimit**: Alarm value for insulation detection, the alarm resistance value for insulation detection; (Unit: Ω/V)
- ak. **IsoErrLimit**: Warning value for insulation detection, the warning resistance value for insulation detection; (Unit: Ω/V)
- al. **StartDelay_s**: Startup delay (s), charging startup delay time.
- am. **preKwh1(kwh)**: Set the starting initial energy;
- an. **preSoc1(%)**: Set the starting initial SOC;
- ao. **preMin1(min)**: Set the starting initial time;
- ap. **TimeChgHour**: Not used temporarily, set to 0;
- aq. **TimeChgMin**: Not used temporarily, set to 0;
- ar. **A_proto**: Not used temporarily, set to 0;
- as. **B_proto**: Not used temporarily, set to 0;

2.3.3.Module Settings

Settings of relevant parameters for the charging module.

Run Config		
Function Sup	Charge Set	Module Set
addrMode		Mode0
MaxTempLimitPower		70.00
PowerLimitCoefficient		0.60
PowerLimitRate		40.00
PowerRealRate		40.00
RateVoltageGrade		1000.00
RateCurrentGrade		40.00
OverVoltage		1001.00
UnderVoltage		100.00
Power_1_StartVol		300.00
Power_1_EndVol		1000.00
Power_2_StartVol		1.00
Power_2_EndVol		1.00
AcSwitchNum		2
DcSwitchNum		1
AcSwOpDealytime(s)		5
BusClDelaytime(s)		10
Temp_Over		100.00
Temp_Bak		5.00
MaxPower		500.00
GunDycWaitTime		15
AcOverVolt		500.00
AcLessVolt		200.00
fanCtrlMode		ChrgContrl
ElcMode		normal
FanLasttime(min)		3
1GroupRmNum		2
2GroupRmNum		2
3GroupRmNum		0
4GroupRmNum		0
5GroupRmNum		0
6GroupRmNum		0
7GroupRmNum		0
8GroupRmNum		0

Figure 2-21 Module Settings Interface

- addrMode: Module address type, 0: Module address starts from 0123, 1: Module address starts from 1234, 2: Module address starts from 1324.
- MaxTempLimitPower: Module over-temperature limit power coefficient, charging module over-temperature limit power coefficient;

- c. PowerLimitCoefficient: Module power limit value, charging module power limit value ($\leq$ module rated power value); (unit: kW)
- d. PowerLimitRate: Module rated power value, charging module rated power value (maximum power value); (unit: kW)
- e. RateVoltageGrade: Module rated voltage grade, charging module rated voltage; (unit: V)
- f. RateCurrentGrade: Module rated current grade, charging module rated current; (unit: V)
- g. overVoltage: Rectifier module output upper limit, charging module output overvoltage threshold; (unit: V)
- h. UnderVoltage: Rectifier module output lower limit, charging module output undervoltage threshold; (unit: V)
- i. Power_1_StartVol: Module constant power 1 segment starting voltage, charging module constant power 1 segment starting voltage; (unit: V)
- j. Power_1_EndVol: Module constant power 1 segment ending voltage, charging module constant power 1 segment ending voltage; (unit: V)
- k. Power_2_StartVol: Module constant power 2 segment starting voltage, charging module constant power 2 segment starting voltage, default is 1V; (unit: V)
- l. power_2_EndVol: Module constant power 2 segment ending voltage, charging module constant power 2 segment ending voltage, default is 1V; (unit: V)
- m. AcSwitchNum: Number of system AC contactors, number of system AC contactors in the equipment;
- n. DcSwitchNum: Number of system bus tie contactors, number of system bus tie contactors in the equipment;
- o. AcSwOpDealytime(s): Delay time for AC contactor disconnection (s), delay time for AC contactor disconnection;
- p. BusCIDelaytime(s): Delay time for bus tie closing (s), delay time for bus tie contactor closing;
- q. Temp_Over: System temperature over-temperature point, equipment ambient temperature over-temperature threshold; (unit: °C)
- r. Temp_Bak: System temperature hysteresis, system temperature hysteresis value; (unit: °C)
- s. MaxPower: System total power limit, equipment maximum output power value; (unit: kW)

- t. GunDycWaitTime: Gun power allocation interval time: interval time for power allocation of each gun in the system; (unit: s)
- u. AcOverVolt: AC overvoltage threshold, AC input overvoltage threshold; (unit: V)
- v. AcLessVolt: AC undervoltage threshold, AC input undervoltage threshold; (unit: V)
- w. fanCtrlMode: Fan control mode, system fan speed control mode (charging control, temperature control, comprehensive control);
- x. ElcMode: System electrical mode, system electrical mode (bus tie scheme: normal mode, ring mode);
- y. FanLasttime(min): Shutdown fan continuous rotation time (min), system fan rotation delay time after stopping charging;
- z. 1GroupRmNum: Number of 1 module configuration, number of modules in the 1st group set;
- aa. 2GroupRmNum: Number of 2 module configuration, number of modules in the 2nd group set;
- ab. 3GroupRmNum: Number of 3 module configuration, number of modules in the 3rd group set;
4GroupRmNum: Number of module configurations in the 4th group, setting for the number of modules in the 4th group;
- ac. 5GroupRmNum: Number of module configurations in the 5th group, setting for the number of modules in the 5th group;
- ad. 6GroupRmNum: Number of module configurations in the 6th group, setting for the number of modules in the 6th group;
- ae. 7GroupRmNum: Number of module configurations in the 7th group, setting for the number of modules in the 7th group;
- af. 8GroupRmNum: Number of module configurations in the 8th group, setting for the number of modules in the 8th group.

2.3.4.Alarm Settings

The fault level settings for the alarm events of the charging station system are as follows: the alarm levels are divided into three grades: shielding, alarm, and fault. The severity of the three levels of events increases successively.

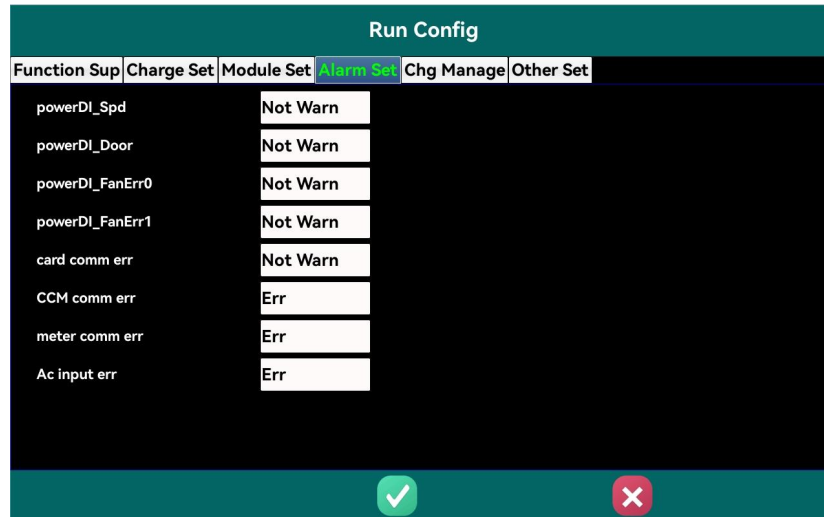


Figure 2-25 Alarm Settings Interface

- a. powerDI_Spd: Lightning arrester failure, setting for the failure level of the lightning arrester; Not Warn; Warn; Err.
- b. powerDI_Door: Door Access Failure: Settings for Door Access Failure Level Not Warn; Warn; Err.
- c. powerDI_FanErrO: Fan failure 1, system fan 1 failure level setting; Not Warn; Warn; Err.
- d. powerDI_FanErr1: Fan failure 2, system fan 2 fault level setting. Not Warn; Warn; Err.
- e. card comm error: Card reader communication failure, No warning; Warn; Err.
- f. CCM error: CCM heartbeat timeout. Not Warn; Warn; Err.
- g. meter comm err: Electric meter communication failure. Not Warn; Warn; Err.
- h. AC input error: Input power failure (overvoltage input voltage fault, undervoltage input voltage fault, missing phase in AC input fault, excessive input current fault, etc.) No warning; Warn; Err.

2.3.5.Charging Management

Settings for charging management-related parameters.

Function Sup	Charge Set	Module Set	Alarm Set	Chg Manage	Other Set
chgCtrlEnable	☐				
chgctrlperiod/s	0				
StartTime-hour	20				
StartTime-min	0				
StopTime-hour	8				
StopTime-min	0				
Trickle-Time/min	60				
Trickle-SOC	90				
Capacity/kWh	200				

Figure 2-26 Charging Management Interface

- chgCtrlEnable: Charging Control Enable Configuration Key;
- chgctrlperiod/s: Charging Control Interval Time Configuration Key/s;
- StartTime-hour: Start Time - Hour;
- StartTime-min: Start Time - Minute;
- StopTime-hour: Stop Time - Hour;
- StopTime-min: Stop Time - Minute;
- Trickle-Time/min: Charging Target Time;
- Trickle-SOC: Charging Target SOC;
- Capacity/kWh: Charging Battery Capacity;

2.3.6. Management of Other Configuration Keys

- GunCertify: Gun parameters - Inspection support configuration key;
- powerCertifys: System parameters - Inspection support configuration key;
- precurBias: Gun parameters - Certification current bias;
- Singlemodule: Gun parameters - Single-module mode support (valid in certification mode);
- CHASStop_30ms: Gun parameters - 30ms shutdown support for target cp3 loss;
- CCSSStop_30ms: Gun parameters - 30ms shutdown support for EU standard cp loss;
- CHA_COMM: Gun parameters - Advanced communication support for target;
- Pre_Pur: Gun parameters - Pre-charging certification circuit support;
- HighVoltSeries: Gun parameters - Module high-voltage series mode support;
- DEKRA_CRT: Gun parameters - DEKRA certification support;
- ISOdeltVStop: Gun parameters - Insulation pressure difference shutdown support;

- l. ISOdelValue: Gun parameters - Insulation detection pressure difference value / V;
- m. GunTempPeriod: Gun parameters - Gun temperature recording period / s;
- n. GunTempSlopStage1: Gun parameters - First-order gun temperature abnormal slope value;
- o. GunTempSlopStage2: Gun parameters - Second-order gun temperature abnormal slope value;
- p. GunTempValueStage2: Gun parameters - Second-order starting gun temperature;
- q. GunTempSuddenValue: Gun parameters - Gun temperature abnormal jump temperature;

2.4.Equipment Information

Query of charging station equipment information: You can view the serial port communication transmission and reception count, IO port feedback, module information, rate information, and version information.

2.4.1.Communication Information

The serial port transmission and reception count between each serial port inside the charging station can be used to determine whether the serial port communication is smooth.

System RunInfo					
Comm Info	IO Feedback	Module State	AcMeter Info	Ver Info	DMU Info
meter1_send		563	meter1_rcv		562
meter2_send		562	meter2_rcv		562
cardRead_send		1448	cardRead_rcv		1448
tcu-ccm1_send		434	tcu-ccm1_rcv		505
tcu-ccm2_send		434	tcu-ccm2_rcv		505
tcu-ccm-srl_send		1624	tcu-ccm-srl_rcv		1623
gun1 connect cnt		0	gun2 connect cnt		0
gun1 sw state cnt		0	gun2 sw state cnt		0

Figure 2-26 Communication Information Interface

- a. meter1_send/rcv: Count sent/received by meter 01, communication between the billing unit (TCU) and the 1st meter;
- b. meter2_send/rcv: Count sent/received by meter 02, communication between the billing unit (TCU) and the 2nd meter;
- c. cardRead_send/rcv: Count sent/received by the card reader, communication between the billing unit (TCU) and the card reader;

- d. tcu-ccm1_send/recv: Count sent/received by TCU-CCM1, communication between the billing unit (TCU) and the 1st main control unit (CCM) via CAN;
- e. tcu-ccm2_send/recv: Count sent/received by TCU-CCM2, communication between the billing unit (TCU) and the 2nd main control unit (CCM) via CAN;
- f. tcu-ccm-srl_send/recv: Count sent/received by TCU-CCM-serial port, communication between the billing unit (TCU) and the main control unit (CCM) via 485 window;
- g. gun connect cnt: (Not available yet);
- h. gun sw state cnt: (Not available yet).

2.4.2.IO Feedback

Viewing of feedback signals from each peripheral of the charging station main control unit (CCM).

System RunInfo					
Comm Info	IO Feedback	Module State	AcMeter Info	Ver Info	DMU Info
gun1_Stop	0		gun2_Stop	0	
gun1_Door	0		gun2_Door	0	
gun1_GunHold	0		gun2_GunHold	0	
gun1_EleLock	0		gun2_EleLock	0	
gun1_Fuse	0		gun2_Fuse	0	
gun1_K1Switch	0		gun2_K1Switch	0	
gun1_WaterOut	0		gun2_WaterOut	0	
gun1_PLCAagvalue	0		gun2_PLCAagvalue	0	
CCM_Stop	0		PCM_Stop	0	
CCM_Door	0		PCM_Door	0	

System RunInfo					
Comm Info	IO Feedback	Module State	AcMeter Info	Ver Info	DMU Info
AC_Switch1	0		AC_Switch2	0	
BusFack1	0		BusFack2	0	
BusFack3	0		BusFack4	0	
BusFack5	0		BusFack6	0	
BusFack7	0		BusFack8	0	
FanFack	0		Light_Protect	0	
In_Breaker	0				
Fan_PWM_1(%)	0		Fan_PWM_2(%)	0	
Fan_PWM_3(%)	0		Fan_PWM_4(%)	0	
Fan_PWM_5(%)	0		Fan_PWM_6(%)	0	

System RunInfo					
Comm Info	IO Feedback	Module State	AcMeter Info	Ver Info	DMU Info
FanFack	0		Light_Protect	0	
In_Breaker	0				
Fan_PWM_1(%)	0		Fan_PWM_2(%)	0	
Fan_PWM_3(%)	0		Fan_PWM_4(%)	0	
Fan_PWM_5(%)	0		Fan_PWM_6(%)	0	
Fan_PWM_7(%)	0		Fan_PWM_8(%)	0	
Fan_PWM_9(%)	0		Fan_PWM_10(%)	0	
Fan_PWM_11(%)	0		Fan_PWM_12(%)	0	
Fan_PWM_13(%)	0		Fan_PWM_14(%)	0	
Fan_PWM_15(%)	0		Fan_PWM_16(%)	0	

Figure 2-27 IO Feedback Interface

- a. gun_Stop: Emergency stop state, feedback status of the emergency stop button (the feedback value indicating the emergency stop button is pressed is 1);
- b. gun_Door: Access control state, feedback status of the access control switch (the feedback value indicating the cabinet door is opened is 1);
- c. gun_GunHold: Gun base, feedback status of the charging gun return position (reserved);
- d. gun_EleLock: Electronic lock feedback, feedback status of the charging gun electronic lock;
- e. gun_Fuse: Fuse feedback, feedback status of the charging gun fuse;
- f. gun_K1SwitchK1: Feedback, DC contactor feedback status;
- g. gun_WaterOut: Water immersion feedback, water immersion fault detection state;
- h. gun_PLCAagvalue: CCM-system emergency stop, main control unit (CCM) detects the feedback status of the emergency stop button;
- i. CCM_Stop: PCM-system emergency stop, power control unit (PCM) detects the feedback status of the emergency stop button;
- j. CCM_Door: CCM-system access control, main control unit (CCM) detects the feedback status of the access control switch;
- k. PCM_Stop: PCM-system access control, power control unit (PCM) detects the feedback status of the access control switch;
- l. Ac_Switch: Feedback of AC contactor 1-2, feedback status of the AC input switch;
- m. BusFack: Feedback of bus 1-8, feedback status of the bus contactor;

- n. FanFack: Fan feedback, fan feedback status;
- o. Light_Protect: Lightning arrester;
- p. In_Breaker: Input circuit breaker feedback, fault feedback status of the input circuit breaker trip.

2.4.3.Module Information

Viewing of status information for the charging station module (AC/DC).

System RunInfo

Comm Info	IO Feedback	Module State	AcMeter Info	Ver Info	DMU Info						
	Group	No.	Output Vol	Output Cur	Temperature	Input Vol_A	Input Vol_B	Input Vol_C	Module State	DC ver	
1	1	1	0.00	0.00	0.00	0.00	0.00	0.00	00000000	0000120a	▲
2	1	2	0.00	0.00	0.00	0.00	0.00	0.00	00000000	00000000	
3	1	3	0.00	0.00	0.00	0.00	0.00	0.00	00000000	00000000	
4	1	4	0.00	0.00	0.00	0.00	0.00	0.00	00000000	00000000	
5	2	1	0.00	0.00	0.00	0.00	0.00	0.00	00000000	00000000	
6	2	2	0.00	0.00	0.00	0.00	0.00	0.00	00000000	00000000	
7	2	3	0.00	0.00	0.00	0.00	0.00	0.00	00000000	0000120f	
8	2	4	0.00	0.00	0.00	0.00	0.00	0.00	00000000	00000000	
9	3	1	0.00	0.00	0.00	0.00	0.00	0.00	00000000	00000000	
10	3	2	0.00	0.00	0.00	0.00	0.00	0.00	00000000	00000000	
11	3	3	0.00	0.00	0.00	0.00	0.00	0.00	00000000	00000000	
12	3	4	0.00	0.00	0.00	0.00	0.00	0.00	00000000	00000000	
13	4	1	0.00	0.00	0.00	0.00	0.00	0.00	00000000	00000000	
14	4	2	0.00	0.00	0.00	0.00	0.00	0.00	00000000	00000000	
15	4	3	0.00	0.00	0.00	0.00	0.00	0.00	00000000	00000000	
16	4	4	0.00	0.00	0.00	0.00	0.00	0.00	00000000	00000000	▼
17											

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1/100

-

×

Figure 2-29 Module Information Interface

- a. Group: Module Group Number, The group number of the module group where this module is located (generally, one charging gun provides charging for one group of modules by default. When other charging module groups are idle, multiple groups of modules will be scheduled to the same charging gun);
- b. No: Serial Number: The number of the module group where this module is located;
- c. Output Vol: Output Voltage, The output voltage of this module;
- d. Output Cur: Output Current, The output current of this module;
- e. Temperature: Environmental Temperature, The environmental temperature sampled and detected by this module;
- f. Input Vol_A: A-phase Voltage, The AC input voltage of phase A;
- g. Input Vol_B: B-phase Voltage, The AC input voltage of phase B;
- h. Input Vol_C: C-phase Voltage, The AC input voltage of phase C;
- i. Module State: Module Information, The alarm status of the module (see Table 3).

2.4.4.Rate

Charging rate query for the charging unit (TCU) of the charging pile.

System RunInfo					
Comm Info	IO Feedback	Module State	AcMeter Info	Ver Info	DMU Info
MeterA_Vol_A(V)		415.32	MeterB_Vol_A(V)	415.90	
MeterA_Vol_B(V)		415.55	MeterB_Vol_B(V)	416.76	
MeterA_Vol_C(V)		416.86	MeterB_Vol_C(V)	415.08	
MeterA_Cur_A(A)		0.00	MeterB_Cur_A(A)	0.00	
MeterA_Cur_B(A)		0.00	MeterB_Cur_B(A)	0.00	
MeterA_Cur_C(A)		0.00	MeterB_Cur_C(A)	0.00	
MeterA_Energy(Wh)		45847	MeterB_Energy(Wh)	59257	
GUNA_Power(kW)		0.00	GUNB_Power(kW)	0.00	

图 2-30 1 交流电表信息显示界面

- a. MeterA_Vol_A(V): Voltage of phase A of gun A.
- b. MeterB_Vol_A(V): Voltage of phase A of gun B.
- c. MeterA_Vol_B(V): Voltage of phase B of gun A.
- d. MeterB_Vol_B(V): External measured voltage of phase B of gun B.
- e. MeterA_Vol_C(v): Internal measured voltage of phase C of gun A.
- f. MeterB_Vol_C(v): External measured voltage of phase C of gun B.
- g. MeterA_Cur_A(v): External measured current of phase A of gun A.
- h. MeterB_Cur_A(v): External measured current of phase A of gun B.
- i. MeterA_Cur_B(v): External measured current of phase B of gun A.
- j. MeterB_Cur_B(v): External measured current of phase B of gun B.
- k. MeterA_Cur_C(v): External measured current of phase C of gun A.
- l. MeterB_Cur_C(v): External measured current of phase C of gun B.
- m. MeterA_Energy(Wh): Charged energy of gun A.
- n. MeterB_Energy(Wh): Charged energy of gun B.
- o. GUNA_Power(kW): Measured output power of gun A.
- p. GUNB_Power(kW): Measured output power of gun B.

2.4.5.Version Information

Query for the software version, hardware version and release date of the charging station.

System RunInfo	
Comm Info	IO Feedback
Module State	AcMeter Info
Ver Info	DMU Info
tcu_softver	02.01.089
tcu_hardver	3.01
tcu_datepublish	2024.03.19
ccm_softver	03.02.026
ccm_hardver	3.10
ccm_datepublish	2024.03.09
IM30	
SECC_A	33240730
SECC_B	34240627
CarInfos-Vendor	
CarInfos-Station	
CarInfos-Version	

Figure 2-32 Version Information Interface

- tcu_softver: TCU software version, the software version number of the charging pile billing unit (TCU);
- tcu_hardver: TCU hardware version, the hardware version number of the charging pile billing unit (TCU);
- tcu_datepublish: The software release date of TCU, the release date of the software of the charging pile billing unit (TCU);
- ccm_softver: CCM software version, the software version number of the charging pile control unit (CCM);
- ccm_hardver: CCM hardware version, the hardware version number of the charging pile control unit (CCM);
- ccm_datepublish: The software release date of CCM, the release date of the software of the charging pile control unit (CCM);
- IM30: IM30 card reader version number (not available yet);
- SECC_A: A conversion board version number;
- SECC_B: B conversion board version number;
- CarInfos-Vendor: Vehicle information Vendor (not available yet);
- CarInfos-Station: Vehicle information Station (not available yet);
- CarInfos-Version: Vehicle information Version (not available yet);

2.4.6.DMU Information

The configuration information of the equipment management unit of the charging station.

System RunInfo

Comm Info	IO Feedback	Module State	AcMeter Info	Ver Info	DMU Info
					DMU Info
1	0	WL_CHAGR/0	ttymxc/0	115200/8/1/0	0/0/50/1/1/17952
2	0	myboard/0	board/0	9600/8/1/0	0/1/10/1/1/676
3	0	DOCA0005ZH/0	ttymxc/1	9600/8/1/2	0/0/10/10/5/172
4	0	DOCA0005ZH	ttymxc/1/ttymxc/1	9600/8/1/2	0/0/10/10/5/172
5	0	card/0	ttymxc/2	115200/8/1/0	0/1/50/1/5/12
6	0	TcuDown/0	can/0	125000/0/0/0	1/0/10/1/1/12
7	0	TcuDown/1	can/0	125000/0/0/0	1/0/10/1/1/12
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					

Figure 2-33 DMU Information Interface

From left to right, they are: DMU ID (default is 0); function name, function number, device name, function number;Baud rate, data bits, stop bits, parity bits; device scheduling mode, parameter source, timeout time, number of receive waits, number of function waits, object memory size.

2.4.7.DCDC/V2G Info

Parameters related to the DCDC/V2G mode

System RunInfo

Comm Info	IO Feedback	Module State	AcMeter Info	Ver Info	DMU Info	DCDC/V2G Info
dcdc/v2g gun1 vol		0.00	dcdc/v2g gun1 cur		0.00	
dcdc/v2g gun2 vol		0.00	dcdc/v2g gun2 cur		0.00	
dcdc gun1 bat vol		0.00	v2g gun1 bat cur		0.00	
dcdc gun2 bat vol		0.00	v2g gun2 bat cur		0.00	
ac avol		0.00	ac acur		0.00	
ac bvol		0.00	ac bcur		0.00	
ac cvol		0.00	ac ccur		0.00	

- dcdc/v2g gun1 vol:1 Voltage on the DC side of the gun 1
- dcdc/v2g gun1 cur:1 Current on the DC side of the gun 1
- dcdc/v2g gun2 vol:2 Voltage on the DC side of the gun 2
- dcdc/v2g gun2 cur:2 Current on the DC side of the gun 2

- e. dcdc gun1 bat vol:1 Voltage on the battery side of the gun 1
- f. dcdc gun1 bat cur:1 Current on the battery side of the gun 1
- g. dcdc gun2 bat vol:2 Voltage on the battery side of the gun 2
- h. dcdc gun2 bat cur:2 Current on the battery side of the gun 2
- i. ac avol: Voltage of output phase A
- j. ac acur: Current of output phase A
- k. ac bvol: Voltage of output phase B
- l. ac bcur: Current of output phase B
- m. ac cvol: Voltage of output phase C
- n. ac ccur: Current of output phase C

2.5. Historical Failures

Viewing of local records for charging station failures.

Histroy Warn		
	Start Time + Code + Gun	
1	2022-09-22 14:12 3032 2	
2	2022-09-22 14:12 3032 1	
3	2022-09-22 09:40 3032 2	
4	2022-09-22 09:40 3032 1	
5	2022-09-21 22:15 3032 2	
6	2022-09-21 22:15 3032 1	
7	2022-09-21 21:19 3032 2	
8	2022-09-21 21:19 3032 1	
9	2022-09-22 01:21 300d 2	
10	2022-09-22 01:21 300d 1	
11	2022-09-22 01:17 300d 2	
12	2022-09-22 01:17 300d 1	
13	2022-09-22 01:16 3002 2	
14	2022-09-22 01:16 3002 1	
15	2022-09-22 01:16 3002 2	
16	2022-09-22 01:16 3002 1	
17	2022-09-22 01:03 3002 2	
18	2022-09-22 01:03 3002 1	
19	2022-09-21 17:00 3002 2	
20	2022-09-21 17:00 3002 1	

Figure 2-33 Historical Fault Interface

The historical fault records consist of three items: the time of the fault occurrence, the fault code, and the fault gun serial number. The maximum number of local fault records that can be displayed is 1000.

2.6. Historical Bills

View local records of charging orders for charging piles.

History Bill								
Number	Order Serial Number	Charge Start Time	Charge Stop Time	SOC(start-end)	Energe(kWh)	Amount	Port	Stop R
1	000000000000...	05-28 15:52	05-28 15:54	50-50	0.003	0.00	1	1003
2	000000000000...	05-28 15:46	05-28 15:49	50-50	0.000	0.00	1	1003
3	000000000000...	05-28 15:29	05-28 15:31	0-0	0.000	0.00	1	5009
4	000000000000...	05-28 15:04	05-28 15:07	0-0	0.000	0.00	1	5009
5	000000000000...	05-28 14:02	05-28 14:03	50-50	0.000	0.00	1	2008
6	512857*****46...	05-27 22:10	05-27 22:11	0-0	0.000	0.00	2	100b
7	512857*****46...	05-27 21:36	05-27 21:37	50-50	0.001	0.00	1	1003
8	512857*****46...	05-27 21:00	05-27 21:01	50-50	0.000	0.00	1	1003
9	512857*****46...	05-27 20:56	05-27 20:58	0-0	0.000	0.00	1	5009
10	512857*****46...	05-27 20:47	05-27 20:49	50-50	0.001	0.00	1	1003
11	512857*****46...	05-27 19:24	05-27 19:26	50-50	0.000	0.00	1	1003
12	512857*****46...	05-27 17:06	05-27 17:07	50-50	0.000	0.00	1	1003
13	512857*****46...	05-27 15:50	05-27 15:51	50-50	0.000	0.00	1	1003
14	512857*****46...	05-27 15:40	05-27 15:41	50-50	0.000	0.00	1	100b
15	512857*****46...	05-27 15:07	05-27 15:09	50-50	0.000	0.00	1	1003
16	512857*****46...	05-27 14:47	05-27 14:49	0-0	0.000	0.00	1	5009
17	512857*****46...	05-27 11:58	05-27 12:00	50-50	0.001	0.00	1	100b

Figure 2-34 Historical Bill Interface

The local history record of charging orders includes: order sequence number, charging start time, charging end time, pre-charging SOC, post-charging SOC, charging electricity quantity, charging amount, charging port and charging stop reason code. The maximum number of historical bill records that can be displayed is 1000.

2.7.Hardware Testing

The main control unit (CCM) of the charging station's main control unit and the external device debugging control interface. (Note: For hardware debugging, control commands need to be issued through the 485 serial port of the gsp background. Therefore, the gsp background communication ports of the TCU and the CCM need to be connected, and then in the parameter settings -> charging settings, the background communication port should be configured as the CCM communication; the current version only supports a single CCM structure and cannot mount multiple terminals.)



图 2-35 硬件测试界面

2.7.1. Switch Control

The status of the interface control buttons is divided into three types:  Indicating that the control status is disconnected,  Indicates that the button has been pressed,  Indicates that the control state is closed.

First, click the button at the top of the page to enter the debugging mode. , When the button status changes and the red button that exits the debugging mode is pressed , The charging station system enters the debugging mode, during which charging is prohibited. Then, the control operation of opening/closing can be carried out by clicking the control buttons behind each switch variable.

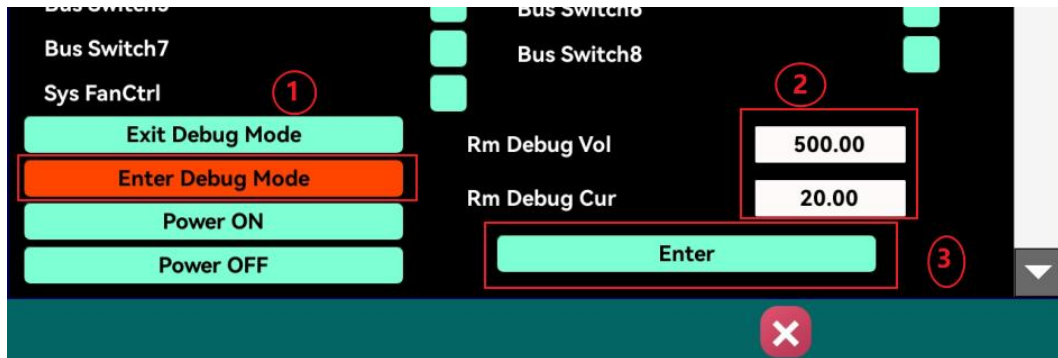
2.7.2.Module Output Control

This test interface also enables module debugging. By pulling the scroll bar all the way down to the bottom, you can see the module debugging function area.

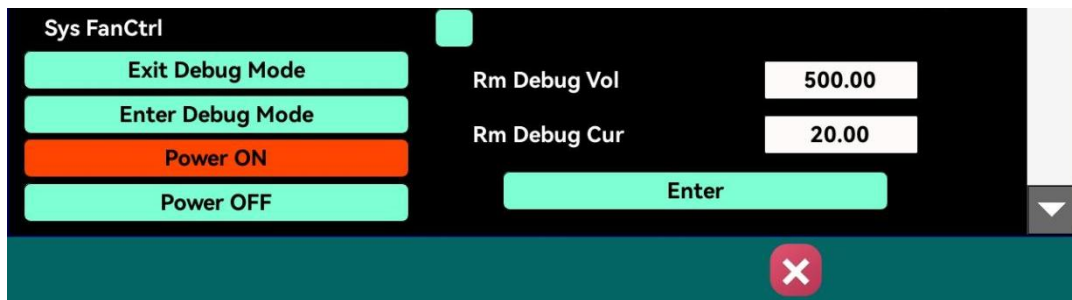


Click the "Debugging Mode" button to enter the module debugging mode. In this mode, charging is prohibited. Click the input box after entering the module's self-start voltage/current,

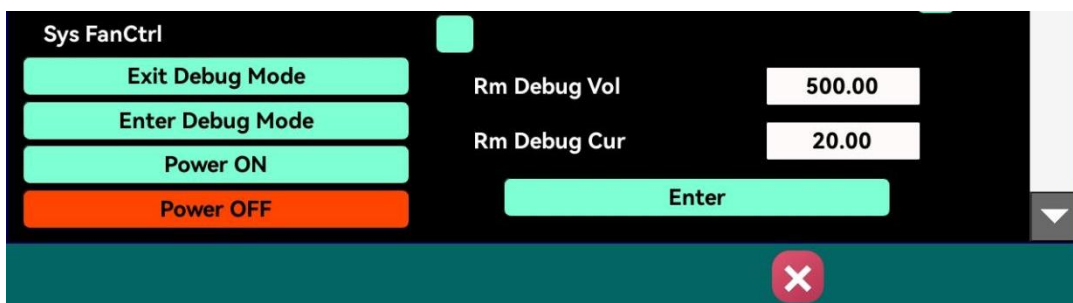
and input the output voltage/current during self-starting. Then click the "OK" button. The voltage/current settings will take effect.



After the self-starting data settings are completed, click the module startup button. All charging modules of the charging pile will start up. After the modules have been started, you can exit this interface and open the module information interface to check the output status of all modules of the charging pile (Menu -> Device Information -> Module Information). This can be used to determine if there are any faulty modules.



Click the module shutdown button, and all modules of the charging station will be shut down. If you need to restart the modules, simply click the module startup button directly.



(Notice: Regardless of whether it is switch debugging or module debugging, after the debugging is completed, the debugging mode must be exited. After exiting the debugging mode, the system will automatically restart.)

2.8.Simulation Charging

The simulation charging debugging interface for the charging station. (Note: For the setting of CCM-related parameters and the issuance of commands, control commands need to be sent through the 485 serial port of the gsp backend. Therefore, the gsp backend communication port of the TCU and the backend communication port of the CCM need to be connected. Then, in the parameter settings -> charging settings, the backend communication port should be configured as the CCM communication; the current version only supports a single CCM structure and cannot mount multiple terminals.)

The simulation charging interface can be divided into three parts: the general settings area ①, the start control area ②, and the charging data area ③.

SimuCharge																											
1 Gun1 Gun2 sup_BMS sup_Guide sup_PreChg sup_TCU simu_Vol (V) 800.00 simu_Cur (A) 125.00 Enter																											
2 simu_auxi 12V simu_inserGun simu_startChg																											
3 <table> <tr> <td>tcu_state</td><td>0</td><td>res_PG</td><td>0.00</td></tr> <tr> <td>tcu_currentErr</td><td>0</td><td>res_NG</td><td>0.00</td></tr> <tr> <td>tcu_stopReason</td><td>0</td><td>meterVol</td><td>0.00</td></tr> <tr> <td>tcu_chgMode</td><td>0</td><td>meterCur</td><td>0.00</td></tr> <tr> <td>meterValue</td><td>2103692</td><td>ccm_state</td><td>0</td></tr> <tr> <td>totalEnergy</td><td>0.00</td><td>ccm_stopReason</td><td>0</td></tr> </table>				tcu_state	0	res_PG	0.00	tcu_currentErr	0	res_NG	0.00	tcu_stopReason	0	meterVol	0.00	tcu_chgMode	0	meterCur	0.00	meterValue	2103692	ccm_state	0	totalEnergy	0.00	ccm_stopReason	0
tcu_state	0	res_PG	0.00																								
tcu_currentErr	0	res_NG	0.00																								
tcu_stopReason	0	meterVol	0.00																								
tcu_chgMode	0	meterCur	0.00																								
meterValue	2103692	ccm_state	0																								
totalEnergy	0.00	ccm_stopReason	0																								

Figure 2-38 Simulated Charging Interface 1

SimuCharge			
meterValue	2103692	ccm_state	0
totalEnergy	0.00	ccm_stopReason	0
chgRuntime	0	ccm_errStopReson	0
needVol	0.00	ccm_currenErr	0
needCur	0.00	currentSoc	0
chg_temp	81	batteryType	0
gun_temp_DC+	30.00	bms_maxVol	0.00
gun_temp_DC-	31.00	bms_maxCur	0.00
busOutVol	0.00	bms_maxTemp	0.00
busInVol	0.00	battery_maxTemp	0.00
busCurrent	0.00	battery_minTemp	0.00
VIN code			

Figure 2-39 Simulated Charging Interface 2

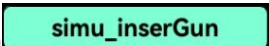
2.8.1.General Settings Area

Click on the buttons labeled "Gun 1" and "Gun 2" in the general area to switch the currently controlled charging gun ports. The buttons will be displayed in red.  Indicates that the current page controls the gun barrel shown by this button.

- a. sup_BMS BMS support: Charging main control unit (CCM) communication function enabling button for vehicle;
- b. sup_Guide Guidance support: Charging gun and vehicle connection guidance voltage detection function enabling button;
- c. sup_PreChg Pre-charge support: Pre-charge function enabling before the official charging of the charging pile;
- d. sup_Tcu TCU support: Charging billing unit (TCU) enabling, after this function is disabled, communication between the charging control unit (CCM) and the charging billing unit (TCU) is disconnected.
- e. These four function support buttons are red  Indicates support; light green indicates no support.
- f. simu_vol (v) Simulation Startup Voltage (V): The output voltage set by the charging control unit (CCM) during a strong charge;
- g. simu_Cur (A) Simulation Startup Current (A): The output current set by the charging control unit (CCM) during a strong charge.

Note: After entering the simulation startup voltage/current, you need to click the "OK" button to enable the input values.

2.8.2.Startup Control Area

- a. "simu_inserGun" - Simulate Insert/Withdraw of Gun: Click the simulated gun insertion button , At this point, the simulated gun-insertion button is transformed into a simulated gun-retrieval button , This charging gun simulates the plug-in state; click the simulated unplug button, and the charging gun state will revert to the simulated unplug state.
- b. simu_startChg Simulate charging/start discharging: Click the simulated charging

button **simu_startChg**, the charging gun initiates the simulation charging process. At this point, this button changes to the simulation stop button **simu_stopChg**. If you need to stop the simulation charging of this gun at this moment, simply click the simulation stop button.

2.8.3. Simulated Startup Charging

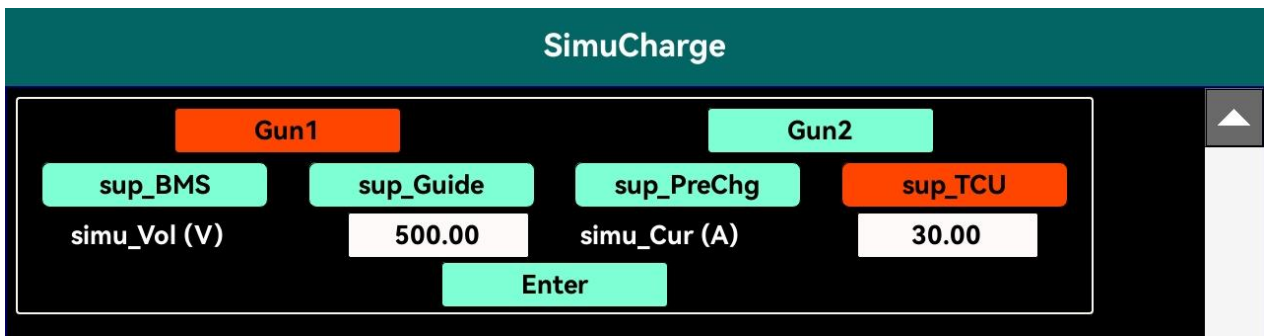
This page supports two methods for initiating charging: through TCU simulation for initiating charging, and through CCM simulation for initiating charging. The main difference between these two methods lies in whether the charging billing unit (TCU) is involved in the charging process.

- 1) Simulated startup charging through TCU: This simulated startup method is initiated by the TCU by sending a startup command. At the same time, the TCU also enters the charging process. Under this startup method, it is necessary to connect the charging gun to the vehicle, which is the same as the user's charging process. No simulated plug-in is required. At this time, the simulated startup voltage and simulated startup current settings are invalid. The actual voltage/current is output according to the vehicle's requirements. At this point, the four supporting items need to be supported to enable normal charging, and there is no need to set the simulated startup voltage/current.

The screenshot shows a user interface for simulated startup charging. It features two columns for 'Gun1' and 'Gun2'. Under 'Gun1', there are buttons for 'sup_BMS' and 'sup_Guide', and a text field for 'simu_Vol (V)' with the value '800.00'. Under 'Gun2', there are buttons for 'sup_PreChg' and 'sup_TCU', and a text field for 'simu_Cur (A)' with the value '125.00'. An 'Enter' button is located at the bottom center.

When the TCU support is enabled, the charging is simulated through the TCU during the startup process.

- 2) Start charging through CCM simulation: This simulation start method directly initiates charging by CCM and does not require TCU to participate in the charging process. Under this start method, there is no need to connect the vehicle for strong charging. The actual voltage/current is based on the set simulation start voltage/current output. Before starting, it is necessary to simulate the plug-in first. When performing strong charging, the three support items "sup_BMS", "sup_Guide", and "sup_PreChg" need to all be changed to not supported (with the orange color changing to light blue).



2.9.Equipment Maintenance

The interface for functions such as software upgrade of the charging unit (TCU) and control unit (CCM), configuration import/export, log export, and restart command operations. (Note: For setting CCM-related parameters and issuing commands, control commands need to be sent through the 485 serial port of the gsp backend. Therefore, the gsp backend communication ports of TCU and CCM need to be connected, and then in the parameter settings -> charging settings, the backend communication port should be configured as the CCM communication. The current version only supports a single CCM structure and cannot mount multiple terminals.)

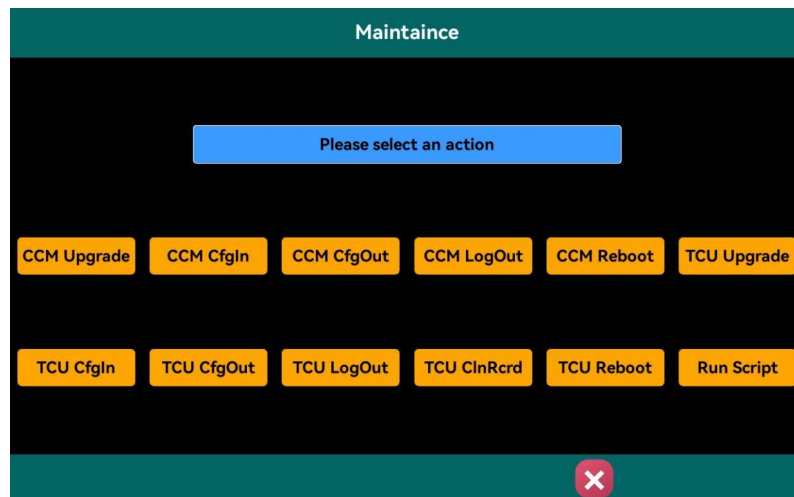


Figure 2-45 Equipment Maintenance Interface

Before conducting the equipment maintenance operation, a new target folder needs to be created in the root directory of the USB drive. (Note: Do not use a USB drive with partitions; otherwise, the software may not be able to recognize the USB drive directory.)

名称	修改日期	类型	大小
cfg	2021/12/25 14:32	文件夹	①
cfqout	2021/12/25 14:32	文件夹	②
logout	2021/12/25 14:33	文件夹	③
upgrade	2021/12/25 14:33	文件夹	④
opt.sh	2021/12/25 14:36	SH 文件	⑤

0 KB

- a. CCM Upgrade: CCM software upgrade, upgrade of the control unit (CCM) software program;
- b. CCM CfgIn: CCM configuration import, import of the configuration file for the control unit (CCM);
- c. CCM CfgOut: CCM configuration export, export of the configuration file for the control unit (CCM);
- d. CCM LogOut: CCM log export, export of the log records of the control unit (CCM);
- e. CCM Reboot: CCM reboot, reboot of the control unit (CCM) program;
- f. Tcu Upgrade: TCU software upgrade, upgrade of the billing unit (TCU) software program;
- g. TCU CfgIn: TCU configuration import, import of the configuration file for the billing unit (TCU);
- h. TCU CfgOut: TCU configuration export, export of the configuration file for the billing unit (TCU);
- i. TCU LogOut: TCU log export, export of the log records of the billing unit (TCU);
- j. TCU ClnRcd: TCU clear record, clear of local fault records and charging records of the billing unit (TCU) (Note: After the clear operation is completed, the TCU needs to be restarted to take effect);
- k. TCU Reboot: TCU reboot, reboot of the billing unit (TCU) program;
- l. Run Script: Run script, execute the script file named opt.sh in the USB drive ⑤.

2.9.1. Software Upgrade Procedure

Before performing the software upgrade, first create a folder named "upgrade" in the root directory of the USB drive. Then, place the software package in this folder. Finally, insert the USB drive into the USB interface of the charging unit (TCU).

名称	修改日期	类型	大小
 ccm.bin ①	2021/10/28 15:35	BIN 文件	537 KB
 tcu_pack ②	2021/10/13 10:33	文件	2,535 KB

- 1) CCM software upgrade: Place the CCM software program in the "upgrade" folder and rename the software name to "ccm.bin" (any other name will cause the software to fail to detect the CCM upgrade software). Then click the CCM program upgrade button and wait for the upgrade to complete. (The CCM program will automatically restart after the upgrade is completed)
- 2) TCU software upgrade: Place the TCU software upgrade package "tcu_pack" in the "upgrade" folder, then click the TCU program upgrade button, and wait for the upgrade to complete. (The TCU program will automatically restart after the upgrade is completed)

2.9.2.Configuration File Import

Before importing the configuration files, first create a folder named "cfg" in the root directory of the USB drive. Then, create two folders named "ccm" and "tcu" respectively in the "cfg" folder, which will be used to store the relevant configuration files of CCM and TCU.

名称	修改日期	类型	大小
 ccm ①	2021/12/25 7:49	文件夹	
 tcu ②	2021/12/25 7:50	文件夹	

- 1) Importing CCM configuration files: The available configuration files that can be imported by CCM are shown in the following figure. The software will sequentially write the files in the order indicated in the figure. It is not necessary for all five configuration files to exist simultaneously. If a certain file is missing, during the burning process, the corresponding file will be skipped. Place these configuration files in the corresponding folder (cfg->ccm), insert the USB drive into the USB interface of the TCU, click the CCM configuration import button, and wait for the burning process to complete. (After the configuration is burned, you need to manually click the CCM restart button to restart the CCM program, so that the new configuration file takes effect.)

名称	修改日期	类型	大小
dev ①	2021/12/25 7:49	Configuration 源...	1 KB
gunlocal ③	2021/12/25 7:49	Configuration 源...	1 KB
gunrecord ⑤	2021/12/25 7:49	Configuration 源...	1 KB
gunuser ④	2021/12/25 7:49	Configuration 源...	4 KB
sys ②	2021/12/25 7:49	Configuration 源...	1 KB

- 2) Import of TCU configuration file: The imported configuration files of TCU are shown in the following figure. Place these configuration files in the corresponding folder (cfg->tcu). Insert the USB drive into the USB interface of TCU, click the TCU configuration import button, and wait for the burning process to complete. (After the configuration burning is completed, you need to manually click the TCU restart button to restart the TCU program, so that the new configuration file takes effect.)

名称	修改日期	类型	大小
cost.user	2021/12/25 7:50	USER 文件	0 KB
devcfg.local	2021/12/25 7:50	LOCAL 文件	0 KB
devComm.user	2021/12/25 7:50	USER 文件	0 KB
local.local	2021/12/25 7:50	LOCAL 文件	0 KB
powerlocal	2021/12/25 7:50	Configuration 源...	0 KB
poweruser	2021/12/25 7:50	Configuration 源...	0 KB
user.user	2021/12/25 7:50	USER 文件	0 KB

2.9.3.Configuration File Export

Before importing the configuration files, first create a folder named "cfgout" in the root directory of the USB drive. Then, create two folders named "ccm" and "tcu" respectively in the "cfgout" folder, which will be used to store the exported CCM and TCU configuration files. Insert the USB drive into the USB interface of the TCU, click the "CCM/TCU Configuration Export" button, and wait for the configuration export operation to complete.

Table 1 Card Type Settings

Card	Card	Remarks
Yonglian	0	Card sector setting 2, sector password: 64 226 1 0 74 60
Cloud	1	Reserved
TUR_CW	2	Singapore card, no key setting, take the physical card
Vega	3	Turkish CW card, no key setting, take the physical card
Hakang	4	Sri Lanka Vega card
Yonglian	5	China Ocean - Hikvision card

表 2 模块告警状态表

Bit	说明
0	Module failure (red indicator light on)
1	Module protection (yellow indicator light on) Maintain
2	Internal SCI communication failure of the module
3	Error in input mode detection (or incorrect input wiring)
4	The monitored input mode does not match the actual working mode Maintain
5	DCDC overvoltage
6	PFC voltage abnormality (unbalanced or overvoltage or undervoltage)
7	AC overvoltage Maintain
8	Maintain
9	Maintain
10	Maintain
11	Communication voltage drop Maintain
12	CAN communication failure
13	Module imbalance current supply Maintain
14	Maintain
15	Maintain
16	Maintain
17	DCDC power-on/off status 0: Power on, 1: Power off

18	Module power limit
19	Temperature power limit
20	AC power limit Maintain
21	Fan failure
22	DCDC short circuit Maintain
23	DCDC over-temperature
24	DCDC input overvoltage
25	Module failure (red indicator light on)
26	Module protection (yellow indicator light on) Maintain
27	Internal SCI communication failure of the module
28	Error in input mode detection (or incorrect input wiring)
29	The monitored input mode does not match the actual working mode Maintain
30	DCDC overvoltage
31	PFC voltage abnormality (unbalanced or overvoltage or undervoltage)